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Hybrid gender agreement in Polish

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Résumé et indexation en français

Accord hybride des genres en polonais
Résumé :
L'objectif de cette thèse est d'étudier l'accord hybride, tel que décrit par Greville Corbett dans ses travaux, en mettant l'accent sur le genre en polonais (Corbett, 1979, 2023). Nous avons mené une étude de corpus en utilisant trois types de noms en polonais pouvant présenter un accord de genre hybride : les noms à genre social, les noms de profession et les formes péjoratives. Sur la base de 1 722 occurrences d'accord avec ces noms trouvées dans le Corpus national de polonais, nous avons constaté que la majorité des occurrences présentaient un accord syntaxique (74 %). Nous avons également constaté des effets de l'ordre cible-contrôleur sur le choix entre l'accord sémantique et l'accord syntaxique dans l'ensemble des données. Nous avons également constaté un effet de la distance dans les occurrences d'accord avec les noms à genre social et un effet du type de cible dans les occurrences avec les noms de profession. Nous avons ensuite mené deux expériences de jugement d'acceptabilité, l'une avec des noms de profession et l'autre avec des formes péjoratives. Nous n'avons pas constaté d'effets significatifs de l'accord sémantique, mais contrairement à l'étude de corpus, nous avons constaté une préférence générale pour l'accord sémantique. Ces études montrent que l'accord sémantique est acceptable en polonais et sensible à plusieurs facteurs de la hiérarchie d'accord de Corbett. Les résultats fournissent également des pistes pour des études futures sur le thème de l'accord de genre hybride en polonais.

Mots clés français : accord ; genre ; polonais ; corpus ; expériences

Forme ou Genre :

fMeSH[1] : Dissertation universitaire

Rameau[2] : Thèses et écrits académiques

[1]Répertoire structuré de mots-clés pour l'analyse de contenu et le classement de documents dans le domaine biomédical (français/anglais)

[2]Répertoire d'autorité-matière encyclopédique et alphabétique unifié (français)

Abstract and indexation in English

Hybrid gender agreement in Polish

Abstract:

The aim of this thesis is to research hybrid agreement, as described by Greville Corbett in his work (Corbett, 1979, 2023), with a focus on gender in Polish. We conducted a corpus study using three types of nouns in Polish that can exhibit hybrid gender agreement: socially gendered nouns, profession nouns and depreciative forms. Based on 1722 agreement occurrences with those nouns found in the National Corpus of Polish we found that the majority of occurrences had syntactic agreement (74%). We also found effects of target-controller ordering on the choice between semantic and syntactic agreement in the whole dataset. We also found an effect of distance in agreement occurrences with socially gendered nouns and an effect of type of target in occurrences with profession nouns. We then conducted two acceptability judgement experiments, one with profession nouns and the other with depreciative forms. We did not find significant effects of semantic agreement, but unlike in the corpus study, we found an overall preference for semantic agreement. The studies show that semantic agreement in acceptable in Polish and sensitive to several factors of Corbett's Agreement Hierarchy. The results also give directions for further studies on the topic of hybrid gender agreement in Polish.

English keywords: agreement; gender; Polish; corpus; experiments

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Introduction

Language can be a surprisingly complex system, especially for something used by people every day with such ease. We generate complex structures with words linked to each other in intricate ways all the time and somehow other people manage to understand the meaning we try to convey through this whole system of communication.

Polish is an example of a language with a relatively free word order, therefore often the links between words are marked morphologically. Agreement between words is one of the ways to figure out if certain words are connected, which means the two words share a value of a certain feature they both have. Though it usually is very simple, sometimes agreement can be extremely convoluted. This study aims to investigate at least some of this complexity to help find ways to describe the agreement patterns and universalities between languages.

Chapter 1

Theoretical background

1.1 Previous work on hybrid agreement

Agreement can be found between a word with some inherent features (a controller) and another, related word (a target). We can see attributive agrrement between a demonstrative and a noun in (1), where the noun is plural, so the demonstrative has to be *these* to agree with the number of the noun. In (2) there is predicate agreement: the subject is the singular pronoun *she*, so the verb has to agree with the number and take the form *reads*. These are examples of agreement marking a syntactic dependency, but it can also be used to mark a discourse dependency. This can be found in (3), where the personal pronoun *she* has to be feminine to have the same referent as the earlier noun phrase *little sister*.

(1) **These dogs** belong to my neighbor.

(2) **She reads** every morning.

(3) I have a little sister, **she**'s 5 years old.

There are cases where the justification for an agreement is rather semantic than syntactic. This can be found in (4): the copular *are* has to be plural, despite the nouns in the subject noun phrase being singular – this is because there is a group reading of the coordinate phrase and there is plural agreement to show that.

(4) John and Mary **are** rich.

Hybrid agreement is possible when there are agreement alternatives, but the controller remains the same. Corbett (2023) provides a simple example of that for number agreement in English: both sentences in (5) are acceptable. The controller of agreement here is the word *family*, the same in both sentences, and there are two agreement alternatives: *have* in (5a) and *has* in (5b). One of them is available, which is known from the lexical semantics of the noun: it can refer to a group of people. The other alternative comes from the morphosyntactic specification: the noun is singular.

(5) a. The family **have** lost everything.
b. The family **has** lost everything.

This is an example of hybrid number agreement, but the same is possible for any other feature that requires agreement in a given language. This study focuses on gender agreement in Polish. Before the system of genders in Polish is described, this section describes Corbett's work on agreement, as it is the main basis for the study conducted later.

Corbett proposed the Agreement Hierarchy, which orders different types of agreement. This hierarchy is used to formulate the constraint visible in (6).

attributive > predicate > relative pronoun > personal pronoun

Figure 1.1: The Agreement Hierarchy (Corbett, 1979)

(6) *Constraint of the Agreement Hierarchy (to be revised)*
For any controller that permits alternative agreements, as we move rightwards along the Agreement Hierarchy, the likelihood of agreement with greater semantic justification will increase monotonically (that is, with no intervening decrease).

We can see this on the example of the word *family*. Starting with the leftmost element of the Agreement Hierarchy, (7a) shows that singular (syntactic) agreement is acceptable, but (7b) shows that plural (semantic) is not. However, with the next step of the hierarchy, predicate agreement, both options are possible, as (5) has shown. Plural agreement is also available with a personal pronoun, which can be found in (8).

(7) a. **This** family **has/have** lost everything.
b. * **These** family **has/have** lost everything.

(8) I saw my family yesterday. **They** seemed to be tired.

The constraint in (6) was proposed for the core instances of hybrid agreement, such as the one shown above (a lexical hybrid), and constructional mismatches, an example of which can be seen in (9).

(9) The girl and the boy **have** been on the playground.

Both nouns in the coordination are singular, but the number in the auxiliary verb *have* agrees with the plural number suggested by the whole construction, instead of the number of the nouns within the construction.

The problem with the Agreement Hierarchy constraint in the form shown above is that it is based on "semantic justification", when hybrid agreement can occur without one of the agreement options having a clear semantic justification. This is clear when we expand the plane of possible hybrid agreements and look at what else can trigger agreement. Alternative sources of agreement information are visible on the left of Table 1.2. The table also gives an idea of where hybrid agreement comes from: it happens whenever different sources give conflicting information about the value of a given feature. This is represented by the thick line which separates the situations where the source suggests syntactic agreement (above the line) from the situations where it suggests semantic agreement (below the line).

		controller type			
		split hybrids	lexical hybrids	constructional mismatch	extraneous
Hierarchy of Agreement Sources	cell specification (1)				
	morphosyntactic specification (2)				
	lexical semantics (3)				
	complex nominal constructions (4)				
	extraneous source (5)				

Figure 1.2: Core (in gray) and generalized semantic agreement (Corbett, 2023, p.8).

This figure includes two more examples of hybrid agreement: the split hybrid *ręka* (meaning 'hand') in Polish and the Norwegian *pancake* sentences with extraneous agreement information.

Polish number system used to have three possible values: singular, dual and plural. Dual is not used anymore, but the word *ręka* (meaning 'hand') is one of few where the relics of dual number can be found: the form used historically as locative dual can now be used as locative singular. In modern Polish the form morphologically resembles masculine or neuter, while the noun itself is feminine. This leads to a situation, where attributive agreement with the locative singular form of the noun can be masculine, as shown in (10). [1]

(10) w **lew-ym** ręk-u
in left-M.SG.LOC hand(F)-SG.LOC
"in the left hand"

There are two conficting sources of agreement: the morphosyntactic specification of the noun is feminine, but the specification of the (singular locative cell) can be

[1] In this example the M1, M2 and M3 forms are syncretic; whenever that is the case, the M label will be used.

masculine/neuter. The question then is: which of the options can be considered "semantic" agreement? One could say the feminine is semantic, because it makes it possible to track the antecedent later in the discourse, but this explanation can be considered to stretch the notion of "semantic" agreement to cover a less-than-semantic example.

On the other hand there is Norwegian "default agreement", that is triggered when there is no straightforward agreement controller, for instance when a sentence contains logical metonymy. As an example Corbett uses the sentence visible below.

(11) Magnus Carlsen og Bobby Fischer hadde vær-t **super-t**.
Magnus Carlsen and Bobby Fischer had be-PST.PTCP **super-N.SG**
"Magnus Carlsen and Bobby Fischer would have been wonderful."

Here, by *Magnus Carlsen and Bobby Fischer* speaker means *having Magnus Carlsen and Bobby Fischer play chess.* Instead of agreeing with the features suggested by the coordination, there is extraneous (default) agreement. Here the conflicting sources of agreement are: coordination of proper nouns that refer to two men and the metonymy, which triggers the "default agreement". Here the example is considered to be more than semantic and therefore not fitting in the core semantic agreement.

Including split hybrids and examples with extraneous agreement information in the analysis allows us to expand beyond semantic agreement and consider agreement options that are less or more than semantic. This is what Corbett did by proposing generalized semantic agreement, along with the revised constraint of Agreement Hierarchy.

(12) *Revised constraint of the Agreement Hierarchy*
For any controller that permits alternative agreements, as we move rightwards along the Agreement Hierarchy, the likelihood of agreement with a greater degree of generalized semantic justification will increase monotonically (that is, with no intervening decrease).

The aim of the current thesis is to see whether the Polish gender agreement pattern really is compatible with the Agreement Hierarchy and with the Hierarchy of Agreement Sources. Polish has a rich gender system, there is gender agreement with adjectives, some verbs and pronouns and it has large lexical and syntactic resources available. First, it is necessary to present the system of grammatical gender in Polish.

1.2 Grammatical gender in Polish

There are multiple options of defining grammatical genders in Polish. The one chosen for this study was proposed by Mańczak (1956), where he differentiates between 5 genders: masculine personal, masculine animate, masculine inanimate, feminine and neuter. The difference between the genders is the easiest to observe when looking at appropriate forms of adjectives in the accusative case. Adjectives agree their gender with nouns and the accusative case provides the most diversity in terms of gender forms.

The example presented in Table 1.3 comes from Mańczak (1956). There are 5 nouns and one adjective for all of them used as an example: *mężczyzna* (meaning 'man'), *pies* (meaning 'dog'), *stół* (meaning 'table'), *kobieta* (meaning 'woman'), *dziecko* (meaning 'child') and *dobry* (meaning 'good').

singular		plural		gender of the noun
adjective	noun	adjective	noun	
dobrego	mężczyznę	dobrych	mężczyzn	masculine personal
	psa	dobre	psy	masculine animate
dobry	stół		stoły	masculine inanimate
dobrą	kobietę		kobiety	feminine
dobre	dziecko		dzieci	neuter

Figure 1.3: Examples of nouns in accusative with 5 grammatical genders in Polish with their agreeing adjectives.

The table shows that each of the nouns agrees with a different set of adjective forms in the accusative singular and plural: *-ego* and *-ych* form for masculine personal, *-ego* and *-e* for masculine animate, *-y* and *-e* for masculine inanimate, *-ą* and *-e* for feminine and *-e* in both numbers for neuter. There are 5 possible sets of agreements, which gives 5 possible grammatical genders.

Saloni (1976) proposed an even more elaborate system, where he talked about 9 genders (instead of one neuter there were two and additionally there were three genders for plurale tantum nouns). Such a system is, however, not necessary for the current analysis, therefore for the sake of simplicity we will adopt the 5 genders proposed by Mańczak. For ease of reference we will use for them the names that Saloni used, which are: M1 (masculine personal), M2 (masculine animate), M3 (masculine inanimate), F (feminine) and N (neuter).

1.3 Polish nouns with possible hybrid gender agreement

Corbett (2023) provides the example of a split hybrid *ręka* (meaning 'hand') in Polish, though unfortunately that example cannot be explored further with the resources available for this study. The circumstances under which the noun has developed to its current use are rare and there are no other nouns which can be analyzed in addition to this one. There are, however, three other categories of nouns which can be used to study hybrid gender agreement and these are described in the following subsections. This work focuses agreement with nouns and does not address complex nominal constructions, such as coordination.

1.3.1 Socially gendered nouns

Here the socially gendered nouns are defined as the nouns that refer to people and semantically involve the social gender or sex of the referent. Corbett (2023) provides the German *das Mädchen* (meaning 'girl') as an example of lexical hybrids – the grammatical gender of the noun is neuter, but the semantics of the word include the feminine gender. The nouns in this group are similar to the German example. There is a number of words in Polish where the gender of the referent is incongruent with the grammatical gender of the noun, which can lead to hybrid agreement. An example, very similar to the German one, can be found in (13).

(13)

To	dziewcz-ę	jest	niemożliw-e!	**Ona**	w	ogóle
this(N.SG.NOM)	girl(N)-SG.NOM	be(PRS.3SG)	impossible-N.SG.NOM	she	in	general

się	nie	uczy!
REFL	NEG	study

"This girl is impossible! She never studies!"

Here the lexical hybrid is essentially the same as the German *das Mädchen* – a noun meaning 'girl' has neuter grammatical gender. In the example we find three words agreeing with the noun: demonstrative *to* agreeing with the grammatical gender, predicate adjective *niemożliwe* agreeing with the grammatical gender and the personal pronoun *ona* in the second sentence agreeing with the semantic gender.

The socially gendered nouns are an instance of lexical hybrids: the conflicting sources of agreement are between the lexical semantics and the morphosyntactic specification of the noun. Most of those nouns refer primarily to women or girls in an expressive (often derogative) manner, but there are different combinations of conflicting values of gender. The process of choosing the nouns for this category and the combinations of conflicting genders are described in the Methodology chapter.

1.3.2 Incongruent profession nouns

In Polish some nouns referring to people holding certain job titles and functions or practising certain professions have masculine gender and there are no feminine nouns with the same meanings. These nouns are likely to be understood to only refer to men, but for a lack of better term they can be also used to refer to women. This can be seen in the examples (14) and (15).

(14)

Now-a	psychiatra	**przepisa-ła**	mi	nowe	leki.
new-F.SG.NOM	psychiatrist(M1)-SG.NOM	prescribed-PST.3SG.F	me	new	medicine

(15)

Now-y	psychiatra	**przepisa-ł**	mi	nowe	leki.
new-M.SG.NOM	psychiatrist(M1)-SG.NOM	prescribed-PST.3SG.M	me	new	medicine

The same noun *psychiatra* is used in both sentences and the only way to tell whether the referent is female or not is by looking at the agreement: feminine in (14) and masculine in (15).

Another option to mark the referent as female is to precede the masculine noun by the word *pani*, which can be translated as 'lady' or 'madam'. It is however possible to omit it.

With recent initiatives to increase womens' visibility in prestige professions and ensure that women are linguistically represented in all fields, there is more debate about those masculine nouns, specifically for which of them it is difficult to create the feminine counterpart and why is it so (Szpyra-Kozłowska, 2019; Szpyra-Kozłowska & Laidler, 2022). The current study takes advantage of this area of linguistic research by using some examples of those masculine nouns as opportunities to find hybrid gender agreement.

1.3.3 Depreciative forms of M1 nouns

In his 1988 paper Saloni points out that for all M1 nouns in plural nominative and vocative case it is possible to create an alternative form, which he calls "depreciative" (Saloni, 1988). This form is created very systematically, specifically by inflecting the noun the same way a noun of a different gender would be inflected. The M1 noun *profesor* (meaning 'professor') will serve as an example here. *Profesor* represents the nominative singular form of the noun. There are two options for the nominative plural form of this noun: *profesorzy* and *profesorowie*, both of these are non-depreciative. To create the depreciative form, it is necessary to look for a noun of a different gender with the same ending. For this example we can use the M3 noun *procesor* (meaning 'processor'), which in plural nominative takes the form *procesory*. Therefore the depreciative form of the noun *profesor* is *profesory*.[2] The example (16) shows the depreciative form used in a sentence, where both attributive agreements and the predicative agreement is non-M1, therefore agreeing with the gender in the specification of the cell.

(16) **Te** **star-e** profesor-y **przysz-ły** w
these(NM1.PL.NOM) old-NM1.PL.NOM professor-PL.NOM.DEPR came-PST.3PL.NM1 in
zabłoconych butach.
muddy shoes.
"These old professors came in with muddy shoes."

Despite the name, the purpose of using a depreciative form is not necessarily to deprecate the referent, although it has been found to be a common function of the form. Based on a corpus study, Siuda (2010) reports that the purpose of 66% depreciative forms is to express negative feelings towards the referent, 24% of the forms are used as a stylistic device, 6% are supposed to reduce the distance to the referent and 4% have a different function.

Depreciative forms can be created only for M1 nouns, but they are created by inflecting them as if they were of any other gender. The conflicting sources of agreement are the following: the morphosyntactic specification of the nominative (and vocative) plural is

[2] In plural nominative and vocative forms are syncretic, therefore the same goes for the vocative form.

M2, M3, N or F, while the morphosyntactic specification of the noun is M1. This makes depreciatives the only form used in this study that represents a different controller type: a split hybrid.

1.4 Hypotheses and research questions

The aim of this study is to investigate the extent to which Corbett's constraints of the agreement hierachy hold on Polish data with hybrid gender agreement. The hypothesis is the following: Polish examples of hybrid gender agreement (socially gendered nouns, profession nouns and depreciative nouns) fit the simple constraint of the agreement hierarchy: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement for all kinds of nouns tested. More semantic agreement is expected in predicate agreement than in attributive and even more in relative pronoun agreement. Furthermore, according to the Hierarchy of Agreement Sources, there should be more semantic agreement with the lexical hybrids (so the socially gendered and profession nouns) than with the split hybrids (depreciative forms). We thus address the following research questions: what is Polish speakers' preference between syntactic and semantic gender agreement? Is this preference sensitive to the type of agreement target or controller?

To test those hypotheses and answer those questions we will conduct a corpus study using socially gendered nouns, profession nouns and depreciative forms and an experimental study using profession nouns and depreciative forms.

Chapter 2

Corpus study

2.1 Methodology

2.1.1 Agreement sources

The corpus used for this study is the 300M subcorpus of the National Corpus of Polish (Przepiórkowski, 2012), which contains texts from various genres and sources, including fiction, journals, letters, Internet, spoken and educational texts, most of which were created after the year 1990. The corpus was parsed using Trankit (Nguyen, Lai, Veyseh, & Nguyen, 2021) to obtain dependency trees conforming to the Universal Dependency standard (UD). Those dependency trees were searched for words, which could trigger hybrid agreement: profession nouns, socially gendered nouns and depreciative forms of m1 nouns. The dependencies of those words then made it possible to search for attributive agreements (adjectives), predicate agreements (verbs agreeing with their subjects) and relative pronoun agreement. A tagged corpus could have been sufficient for attributive and predicate agreement, but using a parsed corpus gave a better chance of finding the correct relations and agreements within them and made it possible to include relative pronouns in the analysis. The subsections below describe how I chose the nouns for analysis.

2.1.1.1 Profession nouns

Nouns in this category came from two studies about feminitives in Polish. The first was a survey reported by Szpyra-Kozłowska (2019), where participants were asked to create feminitives from 50 masculine nouns, that in modern Polish do not have feminine counterparts at all or those counterparts are rarely used. For each noun participants had the option to give no answer. The second study was an experiment conducted by Szpyra-Kozłowska and Laidler (2022), where they tested whether certain feminitives are less accessible by asking participants to try to pronounce them. They chose 20 feminitives, which are formed from masculine nouns that have consonant clusters such as

/kt/ (e.g. *architekt*, meaning 'architect'), /tr/ (e.g. *geriatra*, meaning 'geriatrician') or /rg/ (e.g. *chirurg*, meaning 'surgeon'). According to Klemensiewicz (1957) adding the -ka suffix, the function of which can be to create the feminitve, makes the word difficult to pronounce. Those two studies give a list of 60 nouns in total, for which it is said to be difficult to form feminine counterparts. It is therefore likely that, if there was a female referent of a noun from this list, the masculine noun would be used, leading to hybrid agreement.

Both of the studies determine in some way which of the tested nouns have less available feminitives than others: in Szpyra-Kozłowska and Laidler (2022) those are the nouns for which there are the highest error rates in pronunciation, whereas in Szpyra-Kozłowska (2019) – the nouns for which the most people refused to propose a feminitive. Taking this approach has, however, two downsides: first, it requires an arbitrary threshold that separates the nouns that have an easily formed feminitive from the nouns for which creating the feminine counterpart is more difficult. Second, this approach significantly limits the number of nouns for analysis. Considering this, the current study used all of the nouns from the two studies on Polish feminitves.

Initially, there were 213 070 occurrences of agreement with the 60 nouns in this group. For ease of reference, I call all of the nouns in this category *profession* nouns, despite not all of them actually referring to professions. Meanings of all nouns are presented in Table 2.1. An example of semantic agreement with a profession noun can be seen in (17), where the masculine noun *starosta* (meaning 'mayor') has a feminine adjective *głogowska* (meaning 'of Głogów').

(17) Na zaproszenie starost-y głogowsk-iej Elżbiet-y Urbanowicz
on invitation mayor(M1)-SG.GEN of.Głogów-F.SG.GEN Elżbieta-F.GEN Urbanowicz
- Przysiężn-ej Głogów odwiedziła s. Teresa Jakubowska z
- Przysiężna-F.GEN Głogów visited sister Teresa Jakubowska from
ukraińskiego Żytomierza.
Ukrainian Żytomierz
"Invited by the mayor of Głogów, Elżbieta Urbanowicz-Przysiężna, sister Teresa Jakubowska from Żytomierz in Ukraine visited Głogów."
NCP 300M, sentence ID: 8203800

2.1.1.2 Socially gendered nouns

I chose the nouns for this category using the Polish Academy of Sciences Great Dictionary of Polish available online.[1] With the advanced search option, I searched for nouns in the thematic category HUMAN AS A PHYSICAL BEING: HUMAN PHYSICALITY: SEX. This search yielded 586 results, which then I filtered manually to find the nouns that only refer to humans (and not, for example, abstract nouns or nouns referring to body parts) and that had incongruent grammatical and semantic gender. There were 19 nouns selected this way: 7 of them were grammatically masculine animate and referred to women (e.g. *babsztyl*, meaning

[1] https://wsjp.pl/

Noun in Polish	Translation or definition in English	Noun in Polish	Translation or definition in English
adiunkt	someone who holds a PhD title and is employed at a scientific institution	architekt	architect
burmistrz	mayor	chirurg	surgeon
demiurg	demiurge	dramaturg	playwright
drań	scoundrel	dupek	asshole
dureń	fool	dziekan	dean
ekspert	expert	elekt	person elected in a specified post, but has not yet entered office
foniatra	phoniatrician	ftyzjatra	pulmonologist
geometra	geometer	geriatra	geriatrician
ginekolog	gynecologist	głupek	fool
herszt	ringleader	inżynier	engineer
jaskiniowiec	caveman	jeniec	captive
jeździec	rider	kierowca	driver
ksiądz	priest	lump	tramp
magister	master (holder of a university degree)	majster	foreman
metalurg	metallurgist	minister	minister
murarz	bricklayer	muzykolog	musicologist
mędrzec	wise man	naukowiec	scientist
nieuk	dunce	nurek	diver
pajac	clown	pastor	pastor
pediatra	pediatrician	petent	applicant
polityk	politician	porucznik	lieutenant
powstaniec	insurgent	prefekt	prefect
premier	prime minister	prezydent	president
proboszcz	parish priest	psychiatra	psychiatrist
pułkownik	colonel	rektor	person in charge of a university
sierżant	sargeant	starosta	highest administrative officer of a second-level unit of local government, mayor
subiekt	salesman	szklarz	glazier
szpieg	spy	szubrawiec	rascal
upiór	phantom	widz	audience member
wójt	highest administrative officer of a basic-level unit of local government, mayor	zbieg	fugitive

Table 2.1: Profession nouns (m1) used in the study, with their translations or definitions in English.

'woman' in a pejorative way), 1 was grammatically masculine inanimate and referred to women (*pasztet*, referring to a woman whom the speaker finds ugly or unattractive), 7 were grammatically neuter and referred to women (e.g. *pannisko*, meaning 'a young woman' in an expressive way), 2 were grammatically neuter and referred to men (e.g. *chłopaczysko*, meaning 'a tall or big young man') and 2 were grammatically feminine and referred to men (e.g. *ciota*, meaning 'a homosexual man' in a pejorative way). There were 807 occurrences of agreement with those nouns found in the corpus. The full list of nouns in this category with their translations is in Table 2.2. In (18) there is an example of feminine agreement on the word *ostatnia* (meaning 'last') with the masculine word *babsztyl*.

(18) Ta ostatni-a jest groźn-ym i
this(f.sg.nom) last-f.sg.nom be(prs.3sg) dangerous-m1.sg.ins and
odrażając-ym babsztyl-em
offputting-m1.sg.ins woman-m1.sg.ins
"The last one is a dangerous and offputting woman."
NCP 300M, sentence ID: 12331643

2.1.1.3 Depreciative forms

For this category no list of nouns was created, since the corpus was annotated and I could search for the depreciative forms by checking the words' morphological description. It is possible to create depreciative forms of surnames, however these were excluded from the analysis, because there was no way of determining the referents and therefore no way of ensuring that there is hybrid agreement. Surname *Machała* found in the corpus can serve as an example: let us say that the *Machała* family bought a new car. Assuming the family consists of at least one man, the speaker can use both (19a) and (19b) to convey that information, depending on their attitude they have towards the situation: (19a) if the speaker is neutral or positive and (19b) if they are negative. However, if there are only women in the family, the speaker can only use (19b). Without the context it is unclear whether *Machały* in (19b) is used as a depreciative or simply to refer to women and therefore we cannot decide if this is an example of hybrid agreement.

(19) a. Machał-owie kupi-li nowe auto.
Machała(m1)-pl.nom buy-pst.3pl.m1 new car
b. Machał-y kupi-ły nowe auto.
Machała-pl.nom buy-pst.3pl.nm1 new car
"Machałas bought a new car."

Morphological analyzer Morfeusz (Kieraś & Woliński, 2017) can mark forms as surnames which made it easy to exclude them from the analysis. The other forms marked as depreciative were kept, which gave 254 different lexemes being included as depreciative forms in 1040 agreement occurrences. An example of semantic agreement with a depreciative form can be found in (20), where the noun *chłopy* (meaning 'peasants') in

Noun in Polish	Syntactic gender	Semantic gender	Translation or definition in English
babochłop	M2	feminine	a woman that resembles a man with her physicality, looks and behavior
babol	M2	feminine	a woman that looks or behaves in a way that causes aversion in the speaker
babon	M2	feminine	a woman, about whom the speaker feels negatively
babsko	N	feminine	a woman, about whom the speaker feels negatively
babsztyl	M2	feminine	a woman, about whom the speaker feels negatively
babus	M2	feminine	a woman, about whom the speaker feels negatively
chłopaczysko	N	masculine	a tall or big young man
chłopobaba	F	masculine	a man that resembles a woman with his physicality, looks and behavior (can also mean a woman that resembles a man)
ciota	F	masculine	a homosexual man (can also refer to a woman about whom the speaker feels negatively)
dziewczątko	N	feminine	a girl, whom the speaker likes or to whom they feel compassion
dziewczę	N	feminine	a young girl
dziewuszysko	N	feminine	a girl who causes aversion or contempt in the speaker
kobiecisko	N	feminine	expressively about a woman, augumentative, or to refer to a big woman
kobieciątko	N	feminine	expressively about a woman, diminutive
kobieton	M2	feminine	a woman that resembles a man with her physicality, looks and behavior
pacholę	N	masculine	a boy
pannisko	N	feminine	expressively about a young woman, augumentative
pasztet	M3	feminine	an ugly and unattractive woman
wamp	M2	feminine	a confident, misterious, provocative and attractive woman who seduces men

Table 2.2: Socially gendered nouns used in the study, with their translations or definitions in English.

depreciative form has a non-m1 ending (*chłopy* in contrast to the non-depreciative and clearly masculine *chłopi*), yet still has m1 agreement on the word *pomagać* (meaning 'help').

(20) Chłop-y ze wsi pomaga-li tutaj.
peasants-pl.nom.depr from countryside helped-pst.3pl.m1 here
"Peasants from the countryside helped here."
NCP 300M, sentence ID: 9988833

2.1.2 Extracting agreement occurrences from the corpus

The code used for extracting corpus data in this study can be found in its dedicated GitHub repository.[2] The general idea of how the code works is presented in this section.

2.1.2.1 Finding the agreement controllers

After choosing the nouns to be searched in the corpus, it was necessary to find all of their forms and the forms' morphological descriptions, which I did using the morphological analyzer Morfeusz (Kieraś & Woliński, 2017). First I searched the corpus for occurrences of nouns, which could be controllers of hybrid agreement. Those nouns were saved if they were in the list of gendered or profession noun forms or if they were marked as depreciative in their morphological description. To ensure that no homonyms were accidentaly saved for later analysis, the morphological description of the found word was compared with the one given by Morfeusz. This way, for example, the word *premiera* was excluded if it was an instance of the lexeme *premiera* (meaning 'premiere') in the singular nominative form, but not if it was an instance of the lexeme *premier* (meaning 'prime minister') in the singular accusative form.

Once the right noun was found, I searched the dependency tree for words that can agree their gender with that of the noun. This was the head of the noun (which could be an example of predicate agreement), adjectives attached to the noun (which could be examples of attributive agreement), and relative pronouns referring to the noun. Based on the Universal Dependencies (UD) annotation guidelines, I searched all dependents of the word for a dependency label with a subtype :relcl, which in the UD annotation scheme marks the head of the relative clause. Then I looked for the dependent of the head of the relative clause, that had a UD feature PronType=Rel, which had to be the relative pronoun.[3]

Each occurrence was additionally automatically annotated for the distance between the controller and potential target of agreement (measured in intervening words) and the ordering: whether the target of agreement was the first or not. Once I found the possible agreement targets, I could use the data to look for occurrences of hybrid agreement.

[2]https://github.com/bmagdab/hybrid-agreement

[3]https://universaldependencies.org/workgroups/newdoc/relative_clauses.html

2.1.2.2 Extracting and listing agreement occurrences

To analyse hybrid gender agreement in the found examples, I marked the alternative to the grammatical gender of each agreement controller. For all profession nouns that alternative gender was F, because the only chance of hybrid agreement for those nouns is when the gender of the referent is a woman. For socially gendered nouns the alternative gender was assigned depending on the lexeme and for depreciative forms the alternative gender was always M1, because that is the gender of the lexemes that have depreciative forms.

Once I determined the alternative gender, I excluded plural nouns for which the gender discrepancy was between feminine and neuter. This is because for all types of agreement tested here, the plural feminine and plural neuter forms of targets of agreement are syncretic, therefore there is no hybrid agreement in those cases. This happens, for instance, in (21), where in singular ((21a) and (21b)) there is a difference between the feminine and neuter forms of the word *młoda*. In plural (21c), however, the form is the same for both neuter and feminine, so hybrid agreement cannot be observed. Examples like this were therefore excluded.

(21) a. młod-e dziewcz-ę
young-N.SG.NOM girl(N)-SG.NOM

b. młod-a dziewcz-ę
young-F.SG.NOM girl(N)-SG.NOM
"young girl"

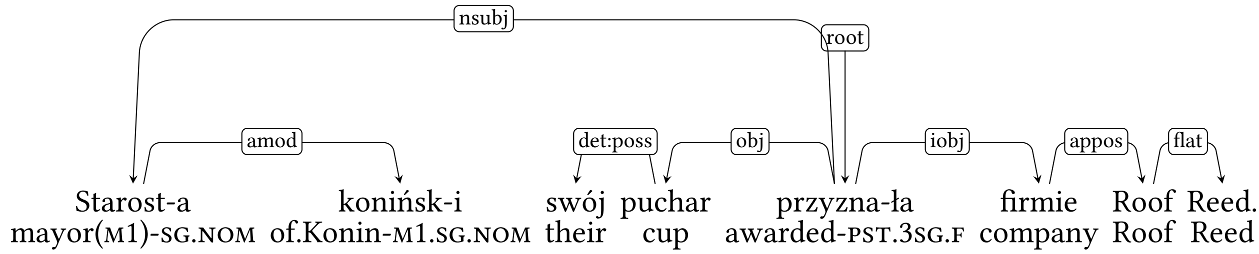
c. młod-e dziewcz-ęta
young-NM1.PL.NOM girl(N)-PL.NOM
"young girls"

The next step was to find occurrences of different types of agreement. The predicate agreement of a given noun was found in its head on three conditions: first, dependency label between the noun and the head had to be NSUBJ, which is the UD label for nominal subjects. Second, the head had to be a verb that inflects for gender, which is only true for verbs annotated as past participles,[4] active and passive participles. Third, the noun had to be in the nominative case. This condition helps filter out situations where a numeral phrase is in subject position, because then the verb does not agree with the subject – it is in 3rd person, singular and neuter form, regardless of the noun. Therefore, predicate agreement cannot be observed in those situations at all and they were excluded.

The search was simpler with attributive and relative pronoun agreement. The only requirement for attributive agreement was that the word was an adjective, which was verified by looking for the tag ADJ in the morphological description of the dependent. As for relative pronouns, all of the ones found earlier were included in the analysis, unless they did not inflect for gender (which was the case, for example, for the word *gdzie* (meaning 'where')).

[4]The annotation scheme used for NKJP uses the past participle label not only for past participles, but also for parts of past tense and conditional mood forms (Przepiórkowski, 2012).

In (2.1) we can see a sentence from the National Corpus of Polish, which serves as an example of the process of extracting agreement described in this chapter. The word, which can trigger hybrid agreement is *starosta* (meaning 'mayor'). There are two words in the sentence that agree with this word in terms of gender: *koniński* (meaning 'of Konin') and *przyznała* (meaning 'awarded'). All of the information saved for each instance of agreement is shown in Table 2.2.



"The mayor of Konin awarded their cup to the company Roof Reed."

Figure 2.1: Dependency tree for the sentence from NCP 300M, sentence ID: 10280659.

		Occurrence 1	Occurrence 2
type of text		typ_publ	
source of agreement	form	starosta	
	lexeme	starosta	
	morphological description	subst:sg:nom:m1	
	grammatical gender	m1	
	other gender	f	
agreement target	form	przyznała	koniński
	morphological description	praet:sg:f:perf	adj:sg:nom:m1:pos
	gender	f	m1
	distance	3	0
	target first	0	0
	semantic agreement	1	0
type of agreement		predicate	attributive

Figure 2.2: All of the information extracted from the sentence in (2.1)

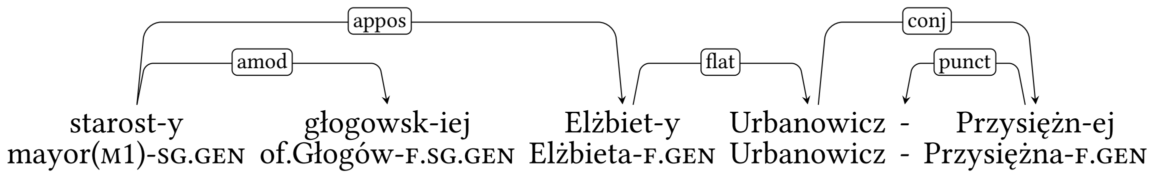
2.1.2.3 Appositions with profession nouns

Initial analysis of agreement occurrences with profession nouns showed, that most occurrences cannot be used in this study, because there is no possibility of determining whether the referent of the profession noun is male or female without the context of the surrounding text (sentences were analysed one by one). Because of this I wrote an additional script to extract information about appositions appearing with profession nouns, because if there is a feminine apposition to the profession noun, it is clear that the referent of the noun is female.[5] This is illustrated by example (17) from a previous section, repeated here. We can be sure that the M1 noun *starosta* is referring to a woman, because the noun phrase is followed by the full name of the mayor in the same case as the noun phrase.

(17) Na zaproszenie starost-y głogowsk-iej Elżbiet-y Urbanowicz
on invitation mayor(M1)-SG.GEN of.Głogów-F.SG.GEN Elżbieta-F.GEN Urbanowicz
- Przysiężn-ej Głogów odwiedziła s. Teresa Jakubowska z
- Przysiężna-F.GEN Głogów visited sister Teresa Jakubowska from
ukraińskiego Żytomierza.
Ukrainian Żytomierz

"Invited by the mayor of Głogów, Elżbieta Urbanowicz-Przysiężna, sister Teresa Jakubowska from Żytomierz in Ukraine visited Głogów."
NCP 300M, sentence ID: 8203800

The process of extracting information about agreement occurrences is almost identical to what was previously described, only here it applied exclusively to profession noun occurrences and had additional steps to find information about appositions. Using the script I first checked if the dependency head of the profession noun found in the corpus was an apposition. If that was the case, all of the dependents and the head of this apposition were treated as the dependents and the head of the profession noun and I looked for gender agreement between them and the profession noun. I calculated the distance again to make sure it is between them and the profession noun and not the apposition and I saved the word form and the morphological description of the apposition, those fields were left empty if no apposition was found. If the head of the noun was not an apposition, the noun's dependants were search to see if any of them was an apposition, in which case the procedure described above was performed. This happened in the sentence in (), and we can see the relevant part of the dependency tree in Figure (2.3).



"mayor of Głogów Elżbieta Urbanowicz-Przysiężna"

Figure 2.3: Dependency tree for a fragment of the sentence in example (17).

[5]The caveat to this approach is that it is not clear whether the feminine agreement with a profession noun is an occurrence of hybrid agreement, or if the agreement is not with the masculine profession noun but with the feminine apposition. This is further discussed in the results section.

2.1.3 Postprocessing of the agreement occurrences

The 214 917 initial observations were filtered to remove those, which were mistakenly included. There were 263 agreement observations with the gendered m3 noun *pasztet*, which as a gendered noun means a girl or a woman whom the speaker finds unattractive, but it is much more common to use that word to refer to a kind of meat. Most of the occurrences found in the corpus were in that meaning, therefore I removed this noun completely from the study.

Some of the occurrences found had to be parsing mistakes, such as large distances between target and controller, reaching up to 1124 intervening words, relative pronouns preceding the controller and targets of agreements being comprised only of digits, which cannot have any gender agreement marking. All of these were removed.

Another group of occurrences which could not be analysed were acronyms, which were marked as depreciative forms. I removed all of them, because analyzing acronyms is outside of the scope of this study. Similar with targets or controllers of agreement being plural, which could mean that they are a part of a coordination. Coordinate structures are a separate step of Corbett's Agreement Hierarchy and also outside of the scope of this study, so I removed them as well.

Some occurrences were suspicious, because they came from a sentence which had surprisingly many agreements (up to 8). Some of them were not actual sentences, but lists of people with certain functions, for instance a list of mayors chosen in local elections in an area. I looked through sentences with unusually many agreements and removed the ones which were actually lists, since any agreement there was a parsing mistake.

In some occurrences the controller was identified by the parser to have the wrong grammatical gender. This could result in inaccurate calculations of semantic agreement, therefore I also went through them to see what should be corrected and what should be discarded. This way some of the occurrences with the noun *zbieg* (meaning 'fugitive') were removed, because it was likely that the non-m1 occurrences of this noun were used in a different meaning, for example in the collocation *zbieg okoliczności* (meaning 'coincidence'). Similarly for the word *upiór* (meaning 'phantom'), which can also be used to mean a haunting memory, for *lump* (meaning 'tramp'), which can also mean old, distressed clothing, for *pajac* (meaning 'clown'), which can also mean a kind of doll and for *nurek* (meaning 'diver'), which can also refer to the act of diving.

The last part of filtering the data concerned the gender of the target of agreement. Possible values of gendered were predefined: m1 or any other with depreciative forms, m1 or f with profession nouns and with gendered nouns it depended on the lexeme. There were some occurrences for which the target of the gender was outside of the possible scope, which was most likely a result of parsing mistakes. For depreciative nouns there was no filtering done in this part, for profession nouns I assumed everything but the m1 or f agreements was parsing mistakes and removed those occurrences, and for gendered nouns I reviewed the incorrectly marked occurrences manually.

2.1.3.1 Additional processing of profession nouns

To figure out whether an occurrence of a profession noun referred to a man or a woman, I added information about appositions attached to the nouns. I removed occurrences with plural and non-F appositions, because the plural appositions were mostly parsing mistakes and the non-F appositions show us that the referent of the noun is not a woman and there cannot be any hybrid agreement, as can be seen in ((22)). Since it is possible that in some cases we agree the target with the gender of the apposition and not the gender of the controller, I marked the ordering of the target, controller and apposition for each agreement to be able to look into the influence of the apposition. In (17) for example, the order was CTA. Only this selection of occurrences with profession nouns was saved for further analysis.

(22)

Dom	nasz	projektowa-ł	mało	znany	architekt-Ø	Stanisław	Gądzikiewicz.
house	our	design-PST.3SG.M1	little	known	architect(M1)-SG.NOM	Stanisław	Gądzikiewicz

"Our house was designed by a little known architect."
NCP 300M, sentence ID: 24

2.2 Results

2.2.1 Overall results

The first hypothesis this study aims to test is: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement. Additionally, the hypothesis proposed with the definition of generalized semantic agreement will be tested, which is that split hybrids (here the depreciative forms) are less likely to have generalized semantic agreement than lexical hybrids (here the socially gendered nouns and profession nouns). Before those hypotheses are addressed, we will look at the dataset extracted from the corpus.

After filtering the data which could not be used, there were 1722 observations, 443 of which were marked as semantic agreement. In Table 2.3 we can see the number of occurrences of agreement for all types of targets (attributes, predicates and relative pronouns) and the percentages of semantic agreement for all those groups.

For statistical modeling of the corpus data I used the `glmer` function from the `lme4` R package and the `glm` function from the `stats` package. I used binomial logistic regression, because the outcome was binary: whether semantic agreement was chosen or not. The first model used all of the 1722 observations. Fixed effects were the type of agreement target (attribute, predicate or relative pronoun), distance between the target and controller, target placement (preceding or following the controller), type of controller (gendered noun, profession noun or depreciative form) and apposition placement (preceding or following the controller). The corpus contains texts from 13 different types

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	891	209	23.46%	682	76.54%
Predicate	686	160	23.32%	526	76.68%
Relative pronoun	145	74	51.03%	71	48.97%
Total	1722	443	25.73%	1279	74.27%

Table 2.3: Numbers and percentages of occurrences of agreement with all nouns.

of sources – this was included as the random effect in the model. The full list of coefficients in this model can be found in the Appendix (Table 4).

Types of targets and the distance were not significant as predictors of semantic agreement. Target being placed after the controller increased the likelihood of semantic agreement (p<0.001), as well as placing an apposition before the controller (p<0.001). Among the noun types the depreciative forms were the baseline, because the hypothesis was that as split hybrids semantic agreement would be less likely to occurr with them than with the lexical hybrids: gendered and profession nouns. Compared to depreciative forms then, socially gendered nouns turned out to be less likely to have semantic agreement (p<0.001), while profession noun were more likely to have semantic agreement (p<0.001).

Figure 2.4 visualises the proportions of semantic and syntactic agreement with different types of agreement, for all nouns combined and for each type of noun separately. The following subsections describe the results for each group of nouns in the study.

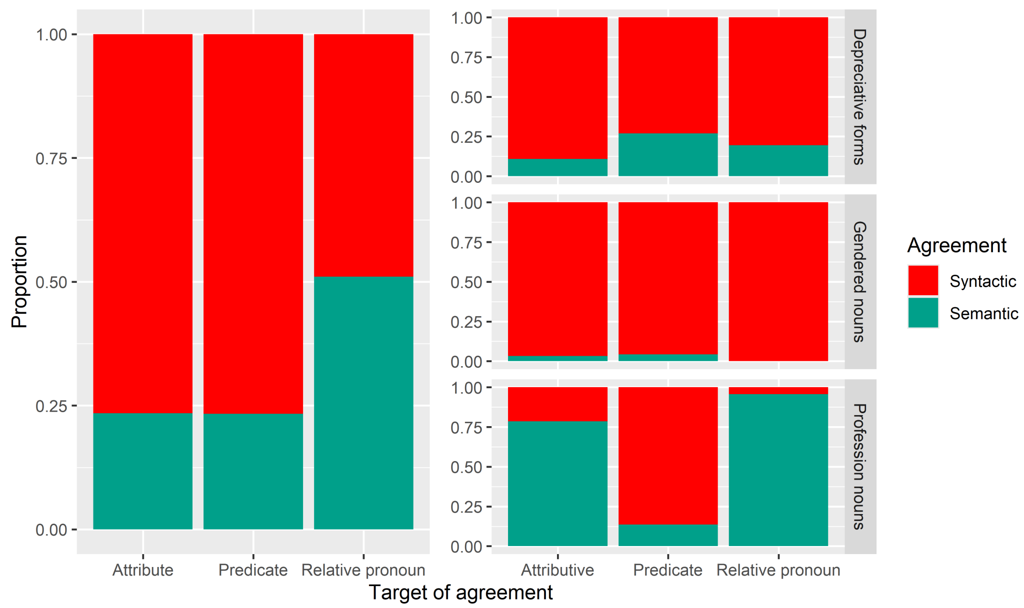


Figure 2.4: Proportions of semantic agreement for each type of target of agreement.

2.2.2 Depreciative forms

Table 2.4 shows the numbers of occurrences and percentages of semantic agreement with depreciative forms. Out of 958 occurrences of agreement with depreciative forms, 20.77% had semantic agreement. The likelihood of semantic agreement was the lowest for attributes (10.79%) and highest for predicates (26.89%), leaving in between the relative pronouns (19.57%). For this type of controller there is no monotonous change of preference for semantic agreement from attributive to relative pronoun agreement, contrary to Corbett's prediction from the Agreement Hierarchy.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	343	37	10.79%	306	89.21%
Predicate	569	153	26.89%	416	73.11%
Relative pronoun	46	9	19.57%	37	80.43%
Total	958	199	20.77%	759	79.23%

Table 2.4: Numbers and percentages of occurrences of agreement with depreciative forms.

In the following examples we can see depreciative forms with all targets of agreement in both possible forms, with the more likely form in bold. For all of them this was the form with syntactic agreement.

(23) a. **Tutej-sze**/tutej-si wykształciuch-y nie
local-NM1.PL.NOM/local-M1.PL.NOM educated.person-PL.NOM.DEPR NEG
są polskimi inteligentami.
be(PRS.3PL) Polish intellectuals
"The local educated people are not the Polish intellectuals."
NCP 300M, sentence ID: 12590744

b. Mieszczuch-y **wol-ały-by**/wol-eli-by, żeby
city.dwellers-PL.NOM.DEPR **prefer-3PL.NM1-COND**/prefer-3PL.M1-COND that
chichotał i mrugał porozumiewawczo.
giggle and wink knowingly
"The city dwellers would prefer him to giggle and wink knowingly."
NCP 300M, sentence ID: 3883941

c. Mnie najbardziej zbijają z nóg modnisi-e,
me most knock off legs dandy-PL.ACC.DEPR
któ-re/któ-rzy uważają się za orginalne.
which-NM1.PL.NOM/-M1.PL.NOM consider REFL as original
"Dandys, who consider themselves original, surprise me the most."
NCP 300M, sentence ID: 6767871

Binomial logistic regression using only the agreement occurrences with depreciative forms with the type of agreement target, distance and target placement as predictors has shown that only target placement had a significant effect on the outcome – target following the controller increased the likelihood of semantic agreement (p<0.001). The full list of coefficients from this model is in the Appendix (Table 7).

2.2.3 Socially gendered nouns

According to Corbett's Source Hierarchy, semantic agreement should be more acceptable with lexical hybrids than with split hybrids, which would mean that socially gendered nouns would have a higher percentage of semantic agreements than depreciative forms. As we can see in Table 2.5, this is not reflected in the collected corpus data, where only 3.2% of occurrences with gendered nouns have semantic agreement. Here the percentages of semantic agreement change monotonously from attributive to relative pronoun agreement, although it is the syntactic agreement that becomes more likely as we go along the Agreement Hierarchy. Considering this, the data from socially gendered nouns fits only the simple constraint of the Agreement Hierarchy (Corbett, 2023).

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	343	11	3.21%	332	96.79%
Predicate	95	4	4.21%	91	95.79%
Relative pronoun	31	0	0%	31	100%
Total	469	15	3.2%	454	96.8%

Table 2.5: Numbers and percentages of occurrences of agreement with socially gendered nouns.

The following examples come from the corpus and show which of the possible agreements are used.

(24) a. Zachichota-ł jak
giggle-PST.3SG.M1 like
stuprocentow-a/stuprocentow-y ciot-a.
hundred.percent-F.SG.NOM/hundred.percent-M1.SG.NOM faggot(F)-SG.NOM
"He giggled like a 100% faggot."
NCP 300M, sentence ID: 18628964

b. Kobiecisk-o **przytulił-o**/przytulił-a się histerycznie do mnie.
woman(N)-SG.NOM **hug-PST.3SG.N**/hug-PST.3SG.F REFL histerically to me

"The woman hugged me histerically."
NCP 300M, sentence ID: 16919204

c. Pamiętasz historię dziewczę-cia, **któr-e**/któr-a uciekło
remember story girl(N)-SG.GEN **which-N.SG.NOM**/which-F.SG.NOM ran.away
z balu?
from dance
"Do you remember the story of the girl, who ran away from the dance?"
NCP 300M, sentence ID: 18852664

Binomial logistic regression using only the agreement occurrences with socially gendered nouns with the type of agreement target, distance and target placement as predictors has shown that distance was the only significant predictor of semantic agreement: larger distance increased the likelihood of semantic agreement with gendered nouns ($p<0.05$). The full list of coefficients from this model can be found in the Appendix (Table 6).

2.2.4 Profession nouns

As we can see in Table 2.6, there were 295 occurrences of agreement with profession nouns, 77.63% of which had semantic agreement. This percentage is larger than for other types of nouns, however this is most likely because all of the profession noun occurrences included in the analysis appeared with feminine appositions, which increase the likelihood of semantic agreement. I will explore this more later when considering the order in which the apposition, controller and target appear in the sentences. With profession nouns it was more likely that semantic agreement would be used in attributes (78.54% of all attributive agreement occurrences) and in relative pronouns (95.59% of all relative pronoun agreement occurrences), but not in predicates (13.64% of all predicate agreement occurrences). This, again, does not follow the simple constraint of the Agreement Hierarchy proposed by Corbett (2023), according to which there should be a monotonic change from attributive to relative pronoun agreement.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	205	161	78.54%	44	21.46%
Predicate	22	3	13.64%	19	86.36%
Relative pronoun	68	65	95.59%	3	4.41%
Total	295	229	77.63%	66	22.37%

Table 2.6: Numbers and percentages of occurrences of agreement with profession nouns.

The agreement preferences found in the corpus are illustrated in the following examples.

(25) a. Większość złożyli w końcówce kadencji
majority submitted in ending term
obecne-go/**obecne-j** burmistrz-Ø Stefanii Sulińskiej.
current-M1.SG.GEN/**current-F.SG.GEN** mayor(M1)-SG.NOM Stefania Sulińska
"The submitted the majority of them at the end of the term of the current mayor, Stefania Sulińska."
NCP 300M, sentence ID: 10793552

b. One są gotowe do adaptacji do roku 2008 –
they are ready for adaptation to year 2008 -
zapewniał-Ø/zapewniał-a minist-er edukacji
assured-PST.3SG.M1/assured-PST.3SG.F minister(M1)-SG.NOM education
Krystyna Łybacka.
Krystyna Łybacka
"They are ready to be adapted before the year 2008 - assured the minister of education Krystyna Łybacka."
NCP 300M, sentence ID: 4040070

c. Felicity Huffman gra panią naukow-iec,
Felicity Huffman plays lady scientist(M1)-SG.NOM
któr-y/**któr-a** w arktycznej stacji zostaje zarażona
which-M1.SG.NOM/**which-F.SG.NOM** in arctic station becomes infected
pasożytem z kosmosu.
parasite from space
"Felicity Huffman plays the scientist, who is infected with a parasite from space in the arctic station."
NCP 300M, sentence ID: 18469406

It is possible that agreement with profession nouns is more likely to be semantic than with other types of nouns, because in those occurrences there were also feminine appositions to the nouns (which were necessary, because without them it would be impossible to tell in corpus examples whether the noun referes to a woman or not). In those situations it is possible that speakers chose feminine agreement on targets (which for profession nouns is semantic agreement), because they were actually agreeing the gender of the target with the gender of the apposition. To have more insight into the structure of those sentences I marked the order in which the appositions (A), controllers (C) and targets (T) appeared with each occurrence of agreement. Figure 2.5 shows the proportions of semantic to syntactic agreement in occurrences with each possible order of appositions, controllers and targets.

Three of the orders shown in the figure noticeably favor semantic agreement over syntactic: apposition-controller-target, apposition-target-controller and target-apposition-controller. In contrast to the other orders, for these the apposition appears before the controller. Since this seems to be the most important information coming from the order variable, for statistical modeling instead of accounting for the

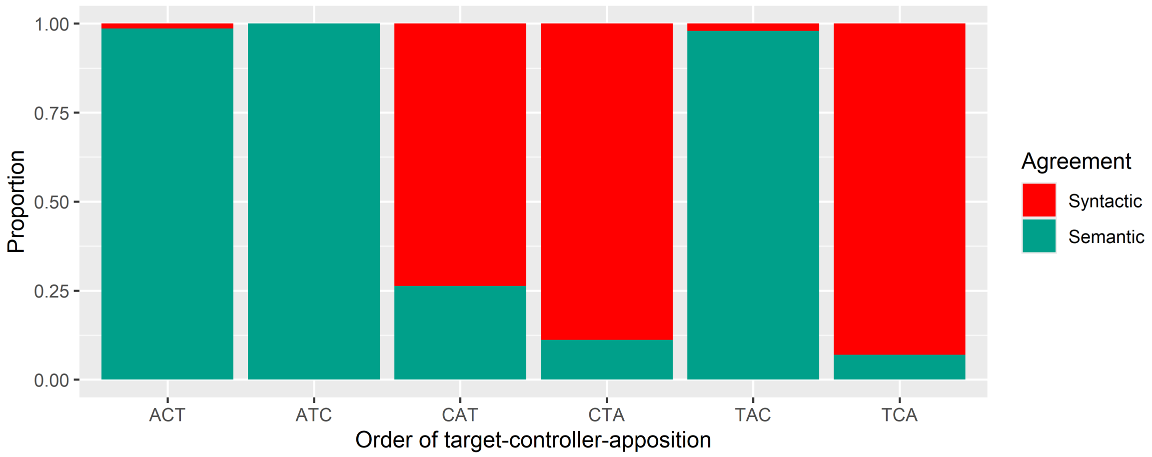


Figure 2.5: Proportion of syntactic and semantic agreements with ordering of the controller, target and apposition

order we will account for whether the apposition appears first (before the controller) or not.

The binomial logistic regression performed on agreement occurrences with profession nouns used the type of agreement target, distance, target placement and apposition placement as predictors of semantic agreement. Distance and target placement were not significant in this model. Contrary to what the Agreement Hierarchy indicates, the odds of attributive agreement being semantic were approximately 16.16 times higher than the odds of predicate agreement being semantic (p<0.05). For relative pronouns the odds were 25.84 times higher than for predicates (p<0.05). There was also a significant effect of apposition placement – the apposition appearing before the controller increased the likelihood of semantic agreement (p<0.001). The full list of coefficients from this model is in the Appendix (Table 5).

Considering the effect of apposition placement on the choice of agreement, we can look at how the proportions of semantic agreement change if we exclude occurrences with apposition first, because the speakers agree the gender of the target with the gender of apposition and not the semantic gender of the controller. Figure 2.6 is identical to Figure 2.4, only with occurrences of profession nouns with appositions first excluded.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	46	3	6.52%	43	93.48%
Predicate	19	3	15.79%	16	84.21%
Relative pronoun	6	3	50%	3	50%
Total	71	9	12.68%	62	87.32%

Table 2.7: Numbers and percentages of occurrences of agreement with profession nouns without occurrences with appositions first.

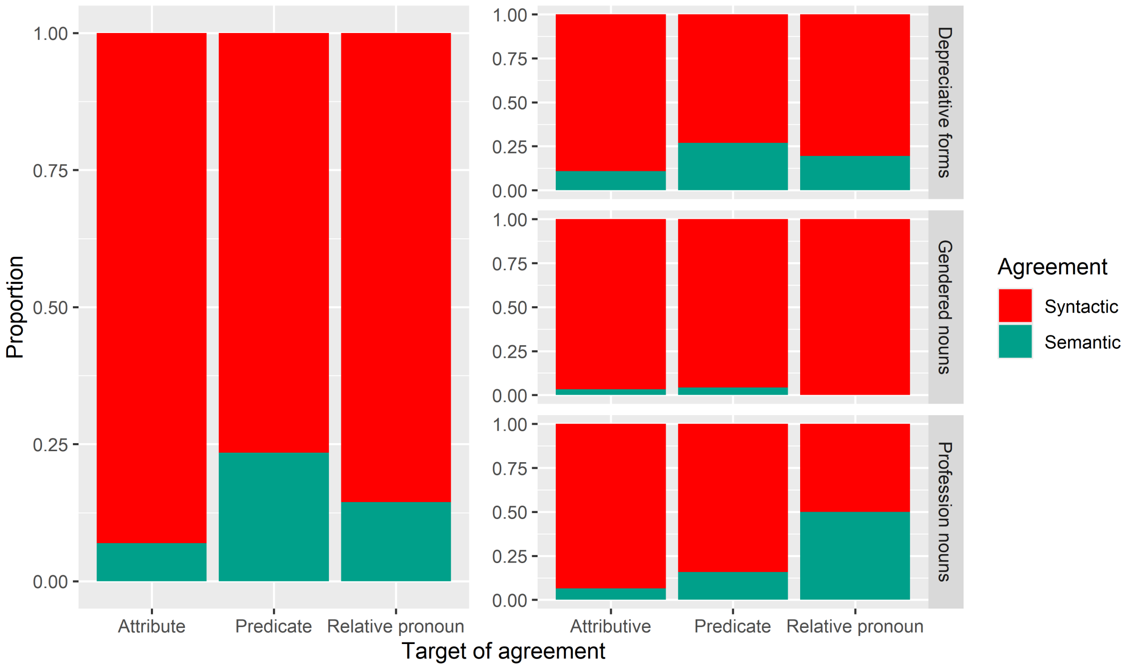


Figure 2.6: Proportions of semantic agreement for each type of target of agreement without occurrences of profession nouns with preceding appositions.

We can see that the overall proportions of semantic agreement (left side of the figure) are now unambiguously not monotonic, and therefore not compatible with any version of the Agreement Hierarchy constraints. The proportions for the profession nouns only, however, now show a monotonic shift from preference for almost exclusively syntactic agreement with attributes, to close to 50% syntactic and semantic agreement.

2.3 Discussion

When looking at all nouns combined, there was a clear preference for syntactic agreement (74% of all occurrences) and no significant effect of the type of target or the distance between the target and the controller. Instead, the choice between semantic and syntactic agreement could be better predicted with the order of target and controller, the type of noun and the apposition – whether it was there and whether it preceded the agreement controller. This is a surprising result, because the type of agreement target was hypothesised to be the main factor in the choice between semantic and syntactic agreement.

As for the effect of the type of noun, based on the Hierarchy of Agreement Sources, the lexical hybrids (profession nouns and socially gendered nouns) should be more likely to have semantic agreement than split hybrids (depreciative forms). There is a significant difference between the lexical hybrids and split hybrids, but it goes in different directions: socially gendered nouns are less likely to have semantic agreement than depreciative forms, while profession nouns are more likely to have semantic agreement than depreciative forms. Looking at this, it is inconclusive whether the lexical hybrids are more likely to have semantic agreement. However, if we exclude the profession nouns with appositions preceding the agreement controllers, we see that only 12.68% of all agreement occurrences are semantic, compared to 20.77% for depreciative forms. It is possible that when appositions precede the controllers, gender of targets is actually agreed with the appositions, and in this case all lexical nouns in this study have less semantic agreement than depreciative forms.

Within the depreciative forms, the only significant factor was the ordering of target and controller: target following the controller increased the likelihood of semantic agreement. This is in line with the hypothesis about the target and controller order.

Among the agreement occurrences with socially gendered nouns there is barely any semantic agreement. One reason for this could be the source of occurrences with those nouns: over 50% of those occurrences come from literature, which can be highly stylized and can affect the choice of agreement with those nouns.

Profession nouns are the only group, for which the type of agreement target was a significant predictor of semantic agreement, but not in the way the Agreement Hierarchy predicts – attributive agreement was more likely to be semantic than predicate agreement. In Figure 2.6 we can see however, that if we exclude the profession noun occurrences with appositions preceding the controller, the proportion of semantic agreement changes monotonically with the smallest in attributive agreement and the largest in relative pronoun agreement. There is the possibility that with apposition preceding the controller of agreement causes speakers to choose the target gender based on the apposition and not on the controller. This seems more likely when we look at the proportions of semantic agreement in Figure 2.5 and take into account that apposition placement was a significant predictor of semantic agreement in both the general and the profession nouns model – for profession nouns only when a feminine apposition appears before the controller it is 63% more likely that there will be semantic agreement, than if the apposition was after the controller.

The main limitation here was that it was a corpus study, therefore it was impossible to control for the variables of interest. There was little data for socially gendered nouns and a lot for profession nouns, which had to be heavily filtered to be included in the analysis. Even the filtering was not fully satisfactory, since the profession noun occurrences had to have appositions included to make sure that there was in fact hybrid agreement. This in turn causes the interpretation of profession noun results to be unclear.

To continue the research on hybrid agreement in Polish, I decided to conduct two acceptability judgement experiments. I decided to use profession nouns and depreciative forms, so that it is possible to reach more definite conclusions regarding the profession nouns, and so that there is more data about split hybrids, which are a newer and less explored addition to the agreement hierarchy. The experimental study allows me to control the context, so that there is no need for using appositions and it is clear that gender agreement on the target comes from the controller.

Chapter 3

Experimental study

3.1 Methodology

We decided to test speakers' preferences for syntactic and semantic agreement in two online experiments: one using one type of lexical hybrids, profession nouns referring to women, the other using split hybrids, depreciative forms which have a form of non-masculine nouns, but refer to men. In the experiments we tested predicative verb agreement and relative pronoun agreement.

I submitted the design of the experiments and the materials to the Paris Graduate School of Linguistics Ethics Committee and got their approval in May 2026. The following subsections describe the experiments in more detail.

3.1.1 Procedure

We conducted two acceptability judgement experiments: one using stimuli with profession nouns and the other with depreciative forms. Both experiments had the same design, which was 3x2 within-subject. The independent variables were the agreement options (semantic or syntactic) and type of agreement with ordering of target and controller: predicate with target first (prenominal), predicate with controller first (postnominal) and relative pronoun (always with controller first, since it is impossible to place the relative pronoun before its antecedent). The dependent variable was the rating of acceptability of given sentences.

The experiment was conducted in Polish. It was recommended to participants to complete the experiment using a computer or tablet, to make sure the instructions and stimuli were displayed correctly. On the first page participants were told their task was to judge whether the displayed sentences sounded naturally or not. They were also informed that participation in the study was voluntary and they could withdraw at any time. To proceed to the study they had to tick a box to confirm they consent. Next, they saw the instructions page, which had the following information.

Your task will be to judge how naturally the displayed sentences sound. If you aren't sure what this means, consider whether you could hear or say the given sentence. You will judge the sentences on a 5-point scale, from "completely unnatural" to "completely natural". You will see a control question after each sentence.

Before we proceed to the study, you will see 3 practice questions that will allow you to familiarize yourself with the task. After the practice session you will proceed to the study, which consists of 60 questions. During the study you cannot go back to the previous questions and there is no time limit to answer the questions, but answer intuitively, without thinking too much.

Out of 60 trials, 24 were with experimental items. They were arranged using the Latin square design – in 6 lists, so that each item appears on a list once and each list has 4 items per condition. Each participant was assigned a list randomly and the order of items was randomized each time. Each trial was followed by a simple comprehension question, to ensure that participants are paying attention to the experiment.

In the experiment on profession nouns the instruction in each trial was "Read the sentences below and judge how naturally the ***second*** one of them sounds", for the experiment on depreciative forms it was "Read the sentence below and judge how naturally it sounds". As was said in the instructions for participants, judgements were given on a 5-point scale. Such a scale is familiar for Polish speakers, because it is similar to the grading scale in Polish schools (1-6 with 6 being the best) and in Polish universities (2-5 with 5 being the best). The scale was labelled only on the ends with *całkowicie nienaturalne* (meaning 'completely unnatural') and *całowicie naturalne* (meaning 'completely natural').

After participants gave judgements on 60 sentences, there was a short demographic survey asking about their age, gender, first language(s) and the country or countries where they were raised. All of the response fields for those questions allowed for any kind of text answer to be typed. The duration of the whole experiment was 15-20 minutes.

3.1.2 Materials

Experimental items were constructed so that the agreement controller was either in the first or the final position in the sentence, the agreement target was immediately adjacent to the controller and there were no other words in the sentence that agreed with the controller in gender. To make sure that the last constraint was not violated, all of the sentences with relative pronoun agreement had to be in present tense (because predicates in present tense do not agree with the subject in gender), while the sentences in other conditions were in past tense. The following subsections describe how the nouns for the experimental items were selected.

3.1.2.1 Experimental items with depreciative forms

Depreciative forms appear in the paradigms of m1 nouns and morphologically they resemble nouns of genders other than m1. In Corbett's approach then, when using those nouns in a sentence we have a syntactic reason to use non-m1 agreement and a semantic reason to use m1 agreement. To make sure that semantic reason is clearly pointing to m1, the nouns selected for this experiment had to be commonly perceived as masculine. To ensure that, the selection process relied on a norming study conducted by Misersky et al. (2014), where they collected information about gender stereotypicality for role nouns in multiple Eurpean languages.

There were 22 nouns that fulfilled the following criteria:

- Perceived to be at least 60% masculine according to the norming study.
- Appear at least a 1000 times in the balanced version of the National Corpus of Polish.
- Depreciative form is not syncretic with the non-depreciative.
- Does not have disturbing, violent connotations (e.g. *rapist*, *murderer*)
- Depreciative form is not syncretic with a different noun (e.g. noun *technik* (meaning 'technician') is *techniki* in depreciative, which has the same shape as the word for 'techniques' and could lead to confusion)
- Is not a nominalized adjective (for which depreciative forms are syncretic with the feminine versions of these nouns)

Additionally, there were two nouns that did not occur in the norming study: *proboszcz* (meaning 'parish priest') and *chłopak* (meaning 'boy'). The former fulfills all of the criteria above, besides being perceived as at least 60% masculine in the norming study, since it is not included there. However, it can be included in the experiment because it is a noun commonly used in Polish and, as a subtype of a priest, it counts as being perceived as masculine.

As Skowrońska (2011) reports, the latter of the two additional nouns is problematic for Polish speakers. In a section dedicated specifically to the lexeme 'chłopak' she reports that some Polish speakers are taught that the non-depreciative form of the noun is incorrect. Because of this, hybrid agreement is very commonly used with this particular noun. The author also looked through the National Corpus of Polish to see a few examples of the noun with attributive and predicate agreement and she finds that predicates with this noun are mostly seen with semantic agreement, while attributes occur almost exclusively with syntactic agreement. This is in line with Corbett's predictions and the current study can test whether this tendency extends to the further points of the Agreement Hierarchy.

All of the nouns chosen for this experiment can be found in the Appendix (Table ?). In (26) there is an example of an experimental item with one of the nouns in depreciative form.

(26) Example of an item in the experiment with depreciative forms in all conditions and with its comprehension question.

a. f/m1, prenominal

Na nagrania do Ameryki Południowej poleci-ały/poleci-eli
for shooting in America South flew-PST.3PL.NM1/flew-PST.3PL.M1
reżyser-y.
directors-PL.NOM.DEPR

"The directors flew to South America for shooting."

b. f/m1, postnominal

Reżysery poleci-ały/poleci-eli na nagrania do
directors-PL.NOM.DEPR flew-PST.3PL.NM1/flew-PST.3PL.M1 for shooting to
Ameryki Południowej.
America South

"The directors flew to South America for shooting."

c. f/m1, relative pronoun

Reżysery, któr-e/któr-zy lecą na nagrania do
directors-PL.NOM.DEPR who-NM1.PL.NOM/who-M1.PL.NOM fly for shooting to
Ameryki Południowej, nie odpowiadają teraz na maile.
America South NEG responding now to emails

"Directors, who are flying to South America to shoot, are not responding to emails right now."

d. comprehension question
Czy w zdaniu wspomniano o Azji?
Did the sentence mention Asia?

3.1.2.2 Experimental items with profession nouns

The process of selecting nouns for this experiment was similar to that for the corpus study. The study by Szpyra-Kozłowska (2019) was used to find masculine nouns, for which it is difficult to use a feminine version. Nouns were arranged from the most times participants of the study refused to give a feminine version of the noun, to the least. This formed a ranking, from which the following nouns were excluded:

- *ksiądz* (meaning 'priest'), *pastor* (meaning 'pastor') and *proboszcz* (meaning 'parish priest'), because in Poland the population of female priests is almost nonexistent
- offensive words (e.g. *dureń* (meaning 'fool'))
- words with less than a 1000 occurrences in the balanced version of the National Corpus of Polish (with literature excluded to make sure the words are used in everyday language)

Here each experimental item also required a sentence of context preceding the target sentence. The nouns used in this experiments are masculine and can refer to both men and women, so context is needed for participants to know that in this situation the noun refers to a woman. All of the items for this experiment can be found in the Appendix (Table ?) and an example of an item can be found in (27).

(27) Example of an item in the experiment with profession nouns in all conditions, with its context and the comprehension question.

a. context
W tym tygodniu odwiedzają nas mamy uczniów z naszej klasy, żeby opowiedzieć o swojej pracy.
Moms of kids from our class are visiting us at school this week to talk about their careers.

b. f/m1, prenominal

We	wtorek	przysz-ła/przysz-edł	architekt-Ø.
on	Tuesday	came-PST.3SG.F/came-PST.3SG.M	architect(M1)-SG.NOM

"The architect came on Tuesday."

c. f/m1, postnominal

Architekt-Ø	przysz-ła/przysz-edł	we	wtorek.
architect(M1)-SG.NOM	came-PST.3SG.F/came-PST.3SG.M	on	Tuesday

"The architect came on Tuesday."

d. f/m1, relative pronoun

Architekt-Ø,	któr-a/któr-y	przyjdzie	we	wtorek,
architect(M1)-SG.NOM	who-F.SG.NOM/who-M.SG.NOM	will.come	on	Tuesday

należy	do	Rady	Rodziców.
belongs	to	council	of.parents

"The architect who will come on Tuesday is a member of the Parents' Council."

e. comprehension question
Czy dzień tygodnia, który był wspomniany w zdaniu, to czwartek?
Was Thursday the day of the week mentioned in the sentence?

3.1.2.3 Other items

Both experiments also had 12 control items, half correct, to establish a threshold of acceptable and unacceptable sentences and test participants' sensitivity to gender agreement. The incorrect sentences contained an attraction error with the subject, the correct ones were the same but without the error. Half of the controls had human subjects and half had non-human, this was distributed evenly within the conditions. In the profession nouns experiment these items were preceded by a sentence of context, so that the experimental item would not be too easy to distinguish. An example of a control item can be found in (28).

(28) Example of a control item in both conditions and the context sentence.

a. context
Nie wszystkie dzieci chciałyby mieć wieczne wakacje.
Not all kids want to be on holidays forever.

b. correct/incorrect

Syn-Ø	sąsiad-ki	jest
son(M1)-SG.NOM	neighbor(F)-SG.GEN	be(PRS.3SG)

szczęśliw-y/*szczęśliw-a	że	idzie	do	szkoły.
happy-M.SG.NOM/*happy-F.SG.NOM	that	goes	to	school

"The son of the neighbor is happy to go to school."

To make sure the main focus of the study is not too obvious to participants, there were also 24 filler items. As with control items there was a context sentence preceding the fillers in the profession noun experiment. An example can be found in (29).

(29) Example of a filler item with its context.

a. context
Szukam zeszytu w którym babcia zapisywała hasła.
I'm looking for the notebook where grandma wrote down her passcodes.

b.

Znalazłem	na	strychu	sejf	i	nie	wiem	jak	go	otworzyć.
found	on	attic	strongbox	and	NEG	know	how	it	open

"I found a strongbox in the attic and I don't know how to open it."

3.1.3 Participants

Participants for both studies were recruited using Prolific. The target population on the platform was filtered to only include people who currently live in Poland, have Polish nationality and speak Polish as their first language. There were in total 60 participants (44 men and 16 women), their age ranged from 18 to 55. They were paid 3.20 GBP (3.7 EUR) for participation. This is 15.69 PLN for 20 minutes which corresponds to 47.07 PLN hourly, so around 150% of minimum hourly wage in Poland as of 2026.

3.1.4 Hypotheses

Based on my corpus study, one hypothesis is that syntactic agreement is preferred over semantic for depreciative forms (possibly for profession nouns too), so items with semantic agreement should have an overall lower acceptability rating than the items with syntactic agreement. The semantic agreement should, however, be rated higher than the ungrammatical control sentences.

Additionally, according to the constraint of Corbett's Agreement Hierarchy, the acceptability of semantic agreement should increase as we go rightwards along the hierarchy. Here this means that semantic predicate agreement has a lower average acceptability rating than semantic relative pronoun agreement. Another hypothesis concerns the Hierarchy of Agreement Sources, based on which semantic agreement with split hybrids is less acceptable than with lexical hybrids. For this study the split hybrids are depreciative forms and lexical hybrids are profession nouns, therefore semantic agreement in the the profession nouns experiment should have an overall higher acceptability than semantic agreement in the depreciatives experiment. Finally, according to Corbett the target of agreement preceding the controller decreases the likelihood of semantic agreement, so in the experiments the prenominal condition should have a lower acceptability than the postnomial condition. In terms of statistical modeling, this means there should be an interaction between agreement and ordering, so that semantic agreement in the prenominal condition is rated lower than in the postnominal condition.

3.2 Results

There were 30 participants invited to participate in each study. Experiment 1 (with profession nouns) had 19 men and 11 women, Experiment 2 (with depreciative forms) had 25 men and 5 women. Participants were aged 18 to 55. Out of them, one responded that he was raised in England and Poland, three reported to speak both Polish and English as their first languages. These participants were excluded from analysis, to make sure the judgements were those of monolingual Polish speakers. All participants responded correctly to comprehension questions over 80% of the time and all of them rated the ungrammatical control questions below the grammatical ones, therefore none of them were excluded based on those criteria. There were, however, two participants in Experiment 2 that rated all experimental items as completely unacceptable. This was presumably because the depreciative forms are highly colloquial and some speakers find those forms to be ungrammatical. Those judgements were therefore not informative for the purposes of this study and responses from those two speakers were excluded. Table 3.1 shows the numbers of participants included in the analysis grouped by gender, age group and experiment.

	Experiment 1			Experiment 2		
	18-30	31-50	51+	18-30	31-50	51+
Women	6	3	1	2	2	0
Men	15	4	0	15	5	1

Table 3.1: Numbers of participants in each age and gender group in each experiment.

The results of both experiments with the control conditions for comparison can be found

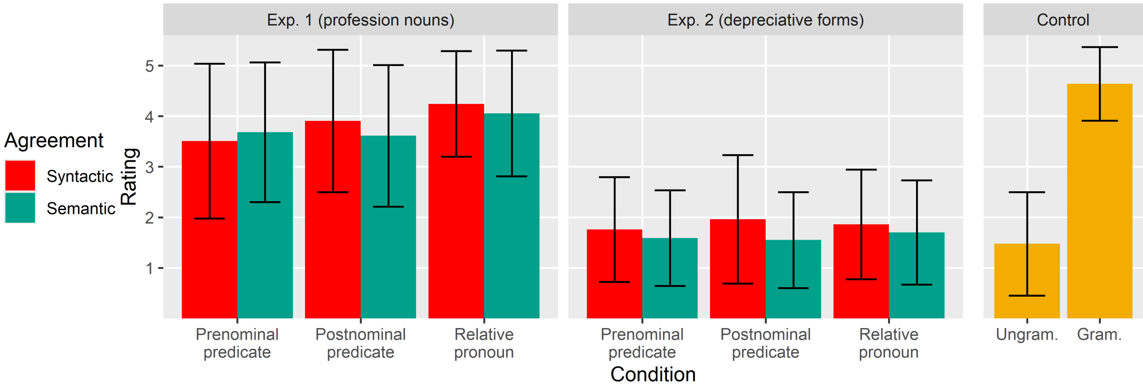


Figure 3.1: Average acceptability judgements for experimental and control conditions in both experiments.

in Figure 3.1. Overall, semantic agreement was rated as high or higher than syntactic agreement. We can see that none of the experimental conditions were on average rated lower than ungrammatical controls, although the items with depreciative forms in Experiment 2 come close to that. We hypothesised that semantic agreement will be less acceptable with depreciative forms (which are split hybrids) than with profession nouns (which are lexical hybrids). While there is a difference between the two types of nouns used, it does not depend on agreement – items with depreciative forms received overall lower ratings than items with profession nouns. There are also differences between ratings of semantic and syntactic agreement in each condition. We can have a closer look at those differences in Figure 3.2. Here we can clearly see that semantic agreement was rated higher than syntactic agreement in all conditions except for prenominal predicates with profession nouns.

Based on the simple constraint of the Agreement Hierarchy (Corbett, 2023) there should be a monotonic change of preference from one value to another. This means that, for instance, syntactic agreement is clearly preferred for attributes, predicate agreement can have both syntactic and semantic agreement and relative pronouns mostly have semantic agreement. In this situation the difference in acceptability of semantic and syntactic agreement for attributes is larger, for predicates smaller, and for relative pronouns larger again (only in a different direction than for attributes). Additionally, there is the full constraint which states that as we go from attributes towards pronouns, that monotonic change of preference should mean an increase in acceptability of semantic agreement (exactly as in the described example).

Here, we can not actually test the monotonicity of change in preference, since only two possible targets from the Agreement Hierarchy were included in the study, but we can look if the differences between preferences for semantic agreement change as expected. Looking at differences for postnominal predicates (which are more canonical than prenominal) and relative pronouns in Figure 3.2 we can see that in both experiments semantic agreement is preferred, because the differences are positive. However, the differences are larger for postnominal predicates than for relative pronouns, which can give us suggestions about the monotonicity of change of preference for one agreement option or the other.

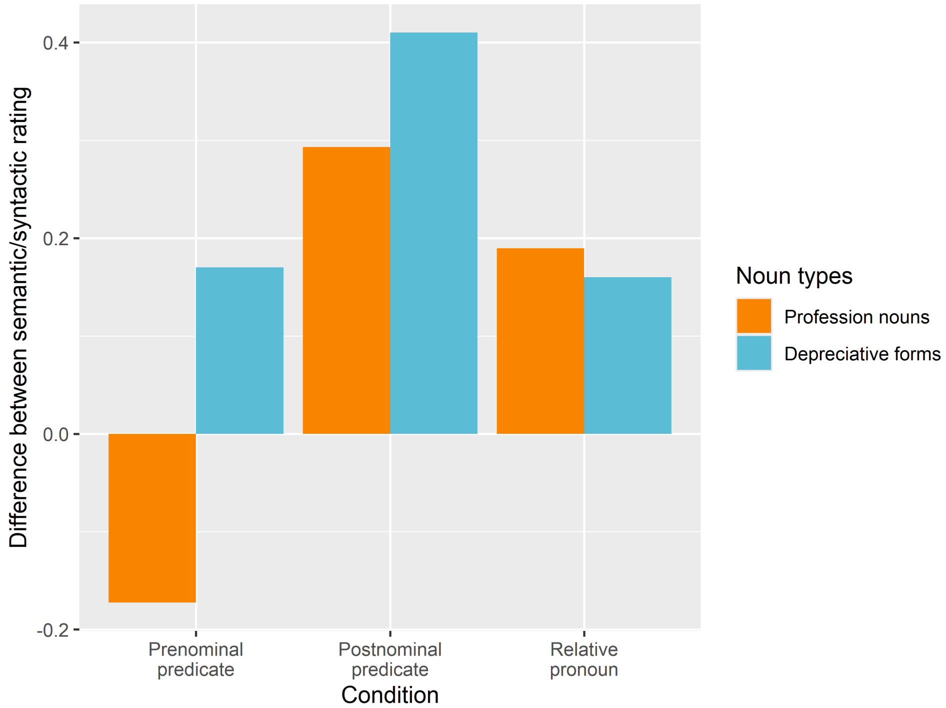


Figure 3.2: Differences between average ratings of semantic and syntactic agreement in both experiments and in all conditions. Positive values mean that semantic agreement was rated higher than syntactic.

We analyzed the results from both experiments with a mixed-effects ordinal regression model using the clmm() function from the ordinal R package (Christensen, 2025). In this model the fixed effects were the noun type (profession noun or depreciative form), agreement (semantic or syntactic), order of target and controller and the type of target (predicate or relative pronoun). The model also included interactions between the noun type and agreement and all other variables. The random effects were for participants, items, order in which items were presented, age and group from the latin square design. The full list of coefficients from this model can be found in Appendix (Table 8).

The only significant effect in this model was the effect of noun: items with profession nouns had significantly higher ratings than items with depreciative forms ($p<0.001$). The following subsections will describe the analyses for the two experiments separately.

3.2.1 Experiment 1

Experiment 1 used stimuli with profession nouns. Table 3.2 shows a summary of the results in this experiment.

For prenominal predicates semantic agreement is rated slightly lower than syntactic, which is expected because target-before-controller ordering should not favor semantic agreement. Then for postnominal predicates the average rating of semantic agreement is higher than for prenominal predicates, which is in line with the hypothesis that semantic agreement

Agreement	Prenominal predicate		Postnominal predicate		Relative pronoun	
	Mean	SD	Mean	SD	Mean	SD
Semantic	3.51	1.53	3.91	1.41	4.24	1.04
Syntactic	3.68	1.38	3.61	1.40	4.05	1.24

Table 3.2: Means and standard deviations for acceptability ratings in all conditions in Experiment 1.

with target first is less acceptable than semantic agreement with controller first. Finally, the relative pronoun agreement has higher ratings than either predicate agreement, which can be in line with the hypothesis derived from the Agreement Hierarchy, but relative pronoun agreement has higher ratings with both semantic and syntactic agreement.

In Figure 3.3 we can see the differences in acceptability depending on the conditions, age group and gender of the participants.

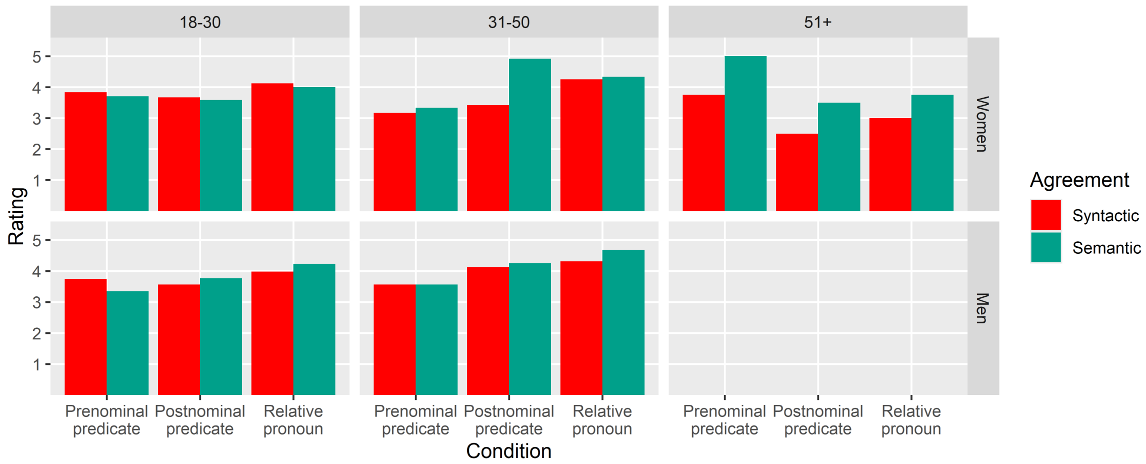


Figure 3.3: Average ratings of experimental items in Experiment 1 grouped by participants' age and gender.

In the plot we can see that women in the youngest group have very similar ratings for all conditions with a slight preference towards syntactic agreement, while the middle group always prefers semantic agreement (it should be noted that the 51+ group here consists of only one person). For men, syntactic prenominal predicate agreement is either the same or better than semantic, while for other conditions they prefer semantic agreement. This is in line with Corbett's prediction that target before controller is more likely to be syntactic.

Results from this experiment were also anaylzed with a mixed-effects ordinal regression model, with agreement crossed with order and target type as predictors of acceptability rating and the same random effects as in the general model. The only significant effect in the model was the type of target: items with relative pronouns had significantly higher acceptability ratings than items with the predicates. However, the interaction between semantic agreement and the type of target was not significant. The list of coefficients from this model can be found in the Appendix (Table 9).

3.2.2 Experiment 2

In Experiment 2 the stimuli had depreciative forms. Table 3.3 summarizes the results of this experiment.

Agreement	Prenominal predicate		Postnominal predicate		Relative pronoun	
	Mean	SD	Mean	SD	Mean	SD
Semantic	1.76	1.04	1.96	1.27	1.86	1.08
Syntactic	1.59	0.94	1.55	0.95	1.70	1.03

Table 3.3: Means and standard deviations for acceptability ratings in all conditions in Experiment 1.

Ratings in Experiment 2 are visibly lower than in Experiment 1, but they never go below the average rating for ungrammatical controls. In all conditions, including prenominal predicates, semantic agreement is surprisingly rated higher than syntactic. Somewhat in line with the hypothesis about target-controller ordering, the preference for semantic agreement over syntactic grows when we move to postnominal predicate condition, where the controller is placed before the target. In contrast to the results from Experiment 1, there is no increase in acceptability of the items with relative pronouns.

Same as for Experiment 1, in Figure 3.4 we can see the average ratings depending on the conditions, age group and gender of the participants.

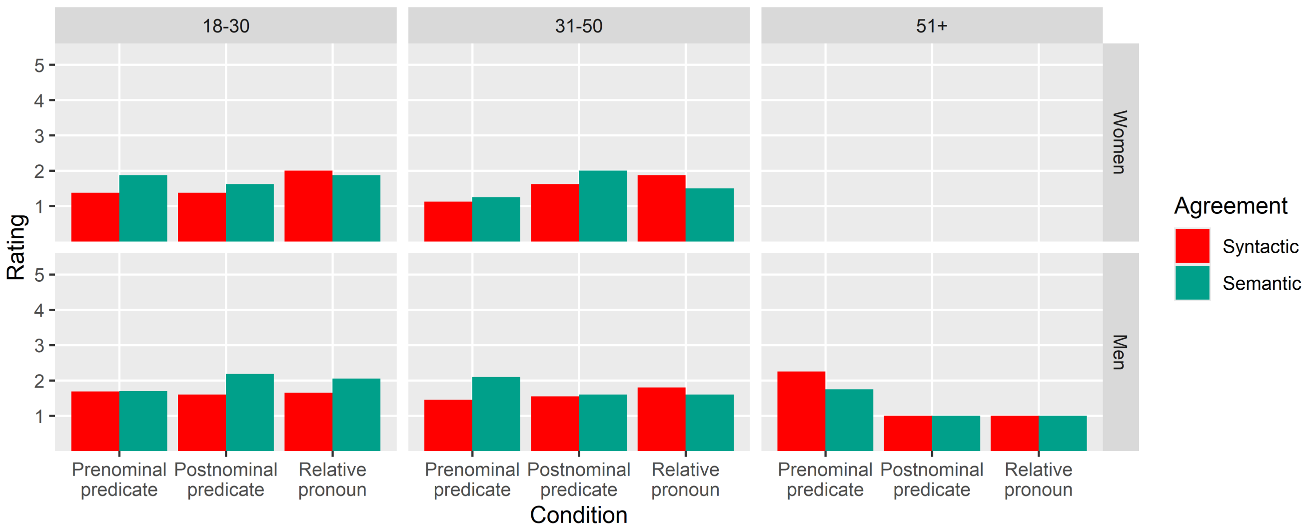


Figure 3.4: Average ratings of experimental items in Experiment 2 grouped by participants' age and gender.

For relative pronouns women in both age groups there is a small preference for syntactic agreement, but for predicates semantic agreement is always rated higher. Men mostly prefer semantic agreement across age groups and conditions, with the exception of relative pronouns for men aged 31-50, where syntactic agreement is rated slightly higher (again, it should be noted that the oldest group consists of only one person here).

We analyzed the data from this experiment with the same mixed-effects ordinal regression model as in the previous section. There were the same fixed effects (agreement, order and target type) and the same random effects. In this model there were no significant effects, neither main effects nor interactions. The list of coefficients for this model is in the Appendix (Table 10).

3.3 Discussion

The only significant effect found in the model with all nouns was that items with depreciative forms were rated significantly lower than items with profession nouns. According to one of the hypotheses, semantic agreement with split hybrids is less acceptable than with lexical hybrids, but here the lower ratings for depreciative forms were independent of agreement: both items with semantic and syntactic agreement were rated as close to unacceptable if they used a depreciative form. There was no significant effect of interaction between agreement and type of noun.

Responses from two participants were filtered out because they rated all depreciative items as completely unacceptable. It seems that those participants thought the depreciative forms are unacceptable themselves and it is possible that this was the case to some extent with most of participants. Depreciative forms are a part of colloquial language and participants could feel that in the experimental settings those forms sound unnatural.

Based solely on descriptive statistics, in the results from Experiment 1 there seem to be some tendencies in the data in line with our hypotheses. On average semantic agreement is more acceptable with postnominal predicates than with prenominal predicates and target following the controller should make semantic agreement more acceptable, but the interaction between agreement and order was not significant. The items with relative pronouns have higher acceptability than items with predicates, and this difference was statistically significant but it was independent of agreement. Since the model showed no significant interaction between agreement and type of target, no conclusions can be drawn regarding acceptability of semantic agreement being higher for relative pronouns than for predicates, as was would be concurrent with the full constraint of the Agreement Hierarchy.

As for depreciative forms, semantic agreement had higher average ratings than syntactic agreement regardless of the condition. Semantic agreement is expected to be less acceptable in general with split hybrids, yet both in the corpus study and in the experimental study was surprisingly acceptable. It is possible that in the experiment, when participants were faced with sentences with “incorrect” forms of nouns, they preferred the version of the sentence with agreement that would be used with the “correct” form of noun. Even if there were some differences between acceptability in different conditions, none of them were statistically significant.

The interesting part of the interaction between the demographic profile of participants and the acceptability ratings in all conditions in Experiment 1 (with profession nouns) is the change in preferences between the youngest women and the group of women aged

31-50. In this experiment there were masculine nouns used for women and with recent efforts to use more feminitives to increase the visibility of women in male-dominated professions, younger women might be more inclined to prefer syntactic agreement with masculine nouns and use feminine agreement with feminine nouns, however uncommon these might be. In a study conducted by Obidzińska (2023), where 300 university students (221 of whom were women) were asked to rate acceptability of feminitive forms, 58.4% of all nouns had the average acceptability of at least 4 on a scale of 0 to 5 and 88.5% were rated at least 3.

Another, slightly surprising, result is the differences between acceptability of semantic and syntactic agreement across conditions (Figure 3.2). Those differences are not significant according to any of the models, but for the sake of future directions it is worth discussing where those results can point. For prenominal predicate agreement the difference is similar for both experiments, but in opposite directions. The more interesting part are the differences in postnominal predicate and relative pronoun conditions. If we expect a monotonic change from syntactic attributive agreement to semantic pronoun agreement, we should see that for the middle points on that scale the difference between acceptability of semantic and syntactic agreement should be smaller than on the ends of the scale. Here, the difference is larger for postnominal predicate agreement and smaller for relative pronoun agreement. It would be beneficial for the future experimental studies to look into semantic and syntactic agreement with all targets on the Agreement Hierarchy.

As for other limitations of the current study, the ratings of items with depreciative forms could be affected by the colloquial nature of the forms themselves. These crucial in research the Agreement Hierarchy extended by generalized semantic agreement, because they are one of rare examples of split hybrids. If these are researched in the future however, it should be emphasised to participants that they should assume the experimental items appear in casual conversations. Another option would be to conduct a similar study, but with audio stimuli, so that participants hear the colloquial forms and are more likely to accept the informal context of the sentences.

The study had relatively little data, considering the number of experimental conditions in the design. Another future direction would be to conduct a similar study on a larger number of participants or with a larger number of items. With more participants there could also be more people in the oldest considered age group, which in the end consisted of only one person per experiment.

It would also be interesting to consider a different experimental design, for example a sentence recall task, where agreement chosen by participants when recalling the experimental items can show which agreement options they prefer.

Conclusion

In the corpus study we found a general preference for syntactic agreement (except for profession nouns). We also found significant effects of target-controller ordering in the whole dataset, an effect of distance for socially gendered nouns and an effect of type of agreement target for profession nouns. The corpus study led to conducting two acceptability judgement experiments: one with split hybrids (depreciative forms) and the other with lexical hybrids (profession nouns). There we found an effect of the type of noun, meaning that items with depreciative forms had significantly lower acceptability ratings than items with profession nouns. There was, however, no interaction with semantic agreement, so there was no definite conclusion regarding the preference for semantic agreement with lexical or split hybrids. In the experiment using profession nouns we also found that items with relative pronouns were significantly more acceptable than items with predicate agreement, but here again there was no interaction with semantic agreement.

The results of the studies show directions for further research: experimental studies with more participants and more types of agreement targets should be conducted to accurately verify the hypotheses regarding hybrid gender agreement. Additionally, agreement with depreciative forms should be studied more, as it is an example of split hybrids, which are a part of a newer extension of the Agreement Hierarchy and are not very common. Studying those forms can give important insight into the rules of hybrid agreement, but it should be done carefully, as these forms are colloquial and speakers may be reluctant to accept them in experimental settings.

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Appendices

A Coefficients from statistical modeling of the corpus data

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-5.99309	0.49265	-12.165	< 2e-16 ***
Attributive agreement	-0.36675	0.22379	-1.639	0.101
Rel. pronoun agreement	-0.34526	0.33058	-1.044	0.296
Distance	0.02303	0.03811	0.604	0.546
Controller before target	0.97644	0.21494	4.543	5.55e-06 ***
Gendered noun	-1.90220	0.28967	-6.567	5.14e-11 ***
Profession noun	1.81877	0.32474	5.601	2.14e-08 ***
Appos. before controller	4.22701	0.42209	10.015	< 2e-16 ***

Table 4: Coefficients of the mixed effects logistic regression model using occurrences with nouns of all types from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-5.1604	1.4187	-3.638	0.000275 ***
Attributive agreement	2.7827	1.2596	2.209	0.027158 *
Rel. pronoun agreement	3.2520	1.3174	2.469	0.013566 *
Distance	0.3212	0.2448	1.312	0.189503
Controller before target	0.9457	0.9044	1.046	0.295722
Appos. before controller	5.7027	0.7131	7.997	1.27e-15 ***

Table 5: Coefficients of the mixed effects logistic regression model using occurrences with profession nouns from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

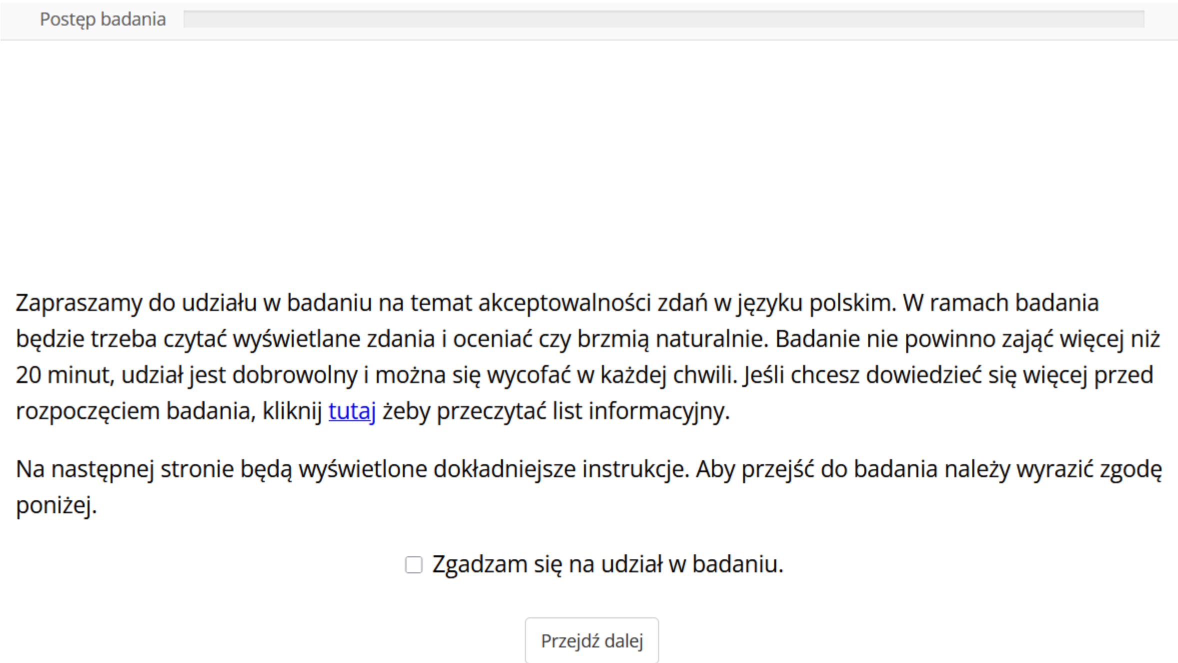
	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-4.6244	0.9319	-4.962	6.97e-07 ***
Attributive agreement	0.8582	0.8527	1.006	0.314223
Rel. pronoun agreement	-14.7725	1124.8988	-0.013	0.989522
Distance	0.4417	0.1220	3.619	0.000295 ***
Controller before target	0.2693	0.6813	0.395	0.692599

Table 6: Coefficients of the mixed effects logistic regression model using occurrences with socially gendered nouns from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Intercept	-1.85889	0.23869	-7.788	6.82e-15 ***
Attributive agreement	-0.47954	0.24973	-1.920	0.0548 .
Rel. pron agreement	-0.61260	0.38700	-1.583	0.1134
Distance	-0.03910	0.04125	-0.948	0.3432
Controller before target	1.07946	0.24116	4.476	7.60e-06 ***

Table 7: Coefficients of the mixed effects logistic regression model using occurrences with depreciative forms from the corpus study.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

B Information for participants in the experiments.



Postęp badania

Zapraszamy do udziału w badaniu na temat akceptowalności zdań w języku polskim. W ramach badania będzie trzeba czytać wyświetlane zdania i oceniać czy brzmią naturalnie. Badanie nie powinno zająć więcej niż 20 minut, udział jest dobrowolny i można się wycofać w każdej chwili. Jeśli chcesz dowiedzieć się więcej przed rozpoczęciem badania, kliknij tutaj żeby przeczytać list informacyjny.

Na następnej stronie będą wyświetlone dokładniejsze instrukcje. Aby przejść do badania należy wyrazić zgodę poniżej.

Zgadzam się na udział w badaniu.

Przejdź dalej

Figure 5: Screenshot of the consent page, the same for both experiments.

Translation of the text on the page:

You are invited to participate in a study about the acceptability of sentences in Polish. As a part of the study you will have to read the displayed sentences and judge whether they sound natural. The study shouldn't take more than 20 minutes, participation is voluntary and you can withdraw at any point. If you want to learn more before beginning the study, click here to read the information letter.

On the next page you will see more specific instructions. To start the study you have to consent below.

I consent to participate in the study.

Continue

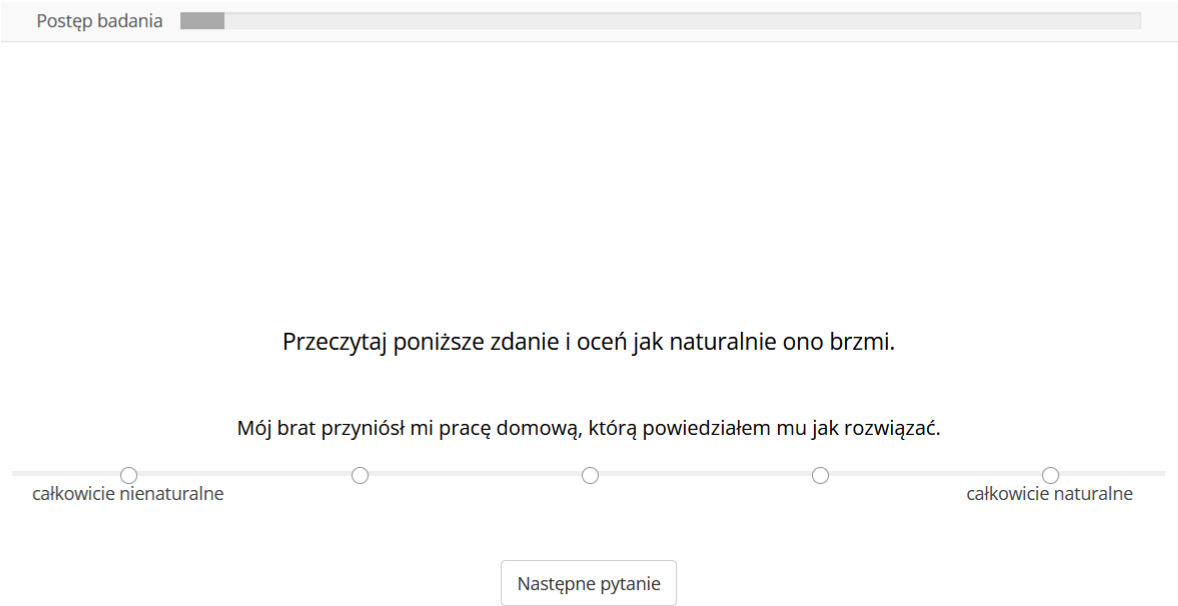


Figure 6: Screenshot of one of the practice items.

Figure 7: Information letter in Polish, the same for both experiments.

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INFORMACJE NA TEMAT UDZIAŁU W BADANIU

Hybrydowe uzgadnianie rodzaju w języku polskim – badanie eksperymentalne

Université Paris Cité, Paris Graduate School of Linguistics, Laboratoire de Linguistique Formelle

Szanowny Panie, Szanowna Pani

Jest Pan/i zaproszony/a do udziału w badaniu prowadzonym przez Magdalenę Borysiak w ramach studiów magisterskich w Paris Graduate School of Linguistics/Laboratoire de Linguistique Formelle na Université Paris Cité.

Ten dokument opisuje badanie, w którym weźmie Pan/i udział, a także odpowiada na pytania które może Pan/Pani mieć.

Zanim zdecyduje się Pan/i wziąć udział w badaniu, istotne jest żeby Pan/i zrozumiał/a cel badania i co się z nim wiąże.

I. Informacje na temat badania

1) Czemu jest Pan/i zaproszony/a do udziału w badaniu?

To badanie zostało Panu/i zaproponowane, ponieważ spełnia Pan/i poniższe kryteria:

- Ma Pan/i ukończone 18 lat
- Polski to Pana/i język ojczysty
- Ma Pan/i obywatelstwo polskie
- Obecnie mieszka Pan/i w Polsce

2) Jakie są cele tego badania?

Celem badania jest dowiedzenie się jakie zdania są akceptowalne dla osób mówiących po polsku.

3) Jakie są korzyści z badania? Czy jest jakieś ryzyko lub ograniczenia powiązane z udziałem w badaniu?

Badanie przyczyni się do naszego rozumienia jak działa język i jak ludzie go używają. Nie ma żadnego ryzyka powiązanego z udziałem w badaniu, przekraczającego to, co spotyka Pan/a w codziennym życiu. Zapewnimy, że wszyscy uczestnicy badania będą traktowani z szacunkiem i rozwagą.

4) Co będzie trzeba zrobić podczas badania?

Podczas badania prezentowane będą różne zdania, a Pana/i zadaniem będzie ocenienie jak naturalnie one brzmią. Jeśli nie jest Pan/i pewien/pewna co to znaczy, proszę zastanowić się czy usłyszenie danego zdania w rozmowie wywołałoby u Pana/i dezorientację. Ocena zdań będzie dokonywana na 5-punktowej skali od „zupełnie nieakceptowalne” do „zupełnie akceptowalne”. Na początku zostaną pokazane 3 pytania ćwiczeniowe, żeby można było oswoić się z zadaniem, po czym będzie można przejść do eksperymentu. Będzie się on składał z 60 pytań, nie będzie można cofać się do poprzednich. Nie ma ograniczenia czasowego na żadne pytanie w eksperymencie, ale zachęcamy do nie zastanawiania się na odpowiedzią zbyt długo i odpowiadania intuicyjnie. Cały eksperyment powinien potrwać około 20 minut.

5) Jakie ma Pan/i prawa jako uczestnik badania?

Może Pan/i zgodzić się lub odmówić udziału w badaniu bez podawania uzasadnienia i bez ponoszenia żadnych konsekwencji.
Może Pan/i poświęcić tyle czasu ile jest potrzebne na podjęcie decyzji.
Jeśli wyrazi Pan/i zgodę na udział, może Pan/i wycofać się w każdej chwili, bez podawania uzasadnienia, kontaktując się przez adres: **magdalena.borysiak@etu.u-paris.fr.**

II. Informacje o przetwarzaniu danych osobowych

6) Czy jakiekolwiek dane osobowe będą przetwarzane w tym badaniu?

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Udział w badaniu wiąże się z przetwarzaniem danych osobowych zebranych i przetworzonych w ramach tego badania.
Dane są przetwarzane wyłącznie w celu badań naukowych na podstawie wykonywania zadania realizowanego w interesie publicznym (art. 6.1.e RODO).
Administratorem danych jest Université Paris Cité.

7) <u>Jakie są Pana/i prawa odnośnie danych osobowych?</u>

Może Pan/i poprosić o dostęp do danych i o ich usunięcie. Ma Pan/i także prawo sprzeciwić się, skorygować i ograniczyć przetwarzanie swoich danych, ale tylko do czasu pełnego zanonimizowania danych.
Żeby skorzystać z tych praw lub dowiedzieć się więcej na temat prowadzonego badania może Pan/i skontaktować się z głównym badaczem pod adresem magdalena.borysiak@etu.u-paris.fr. Odpowiedź otrzyma Pan/i jak najszybciej, nie później niż miesiąc od dostarczenia wiadomości.
W przypadku problemów może Pan/i również skontaktować się z Inspektorem Ochrony Danych na Université Paris Cité pod adresem dpo@u-paris.fr. Jeśli po skontaktowaniu się z wyżej wymienionymi osobami uważa Pan/i, że prawa « Informatique et Libertés » nie są przestrzegane, może Pan/i przesłać skargę do CNIL (www.cnil.fr).

8) <u>W jaki sposób zarządzamy Pana/i danymi osobowymi?</u>

W ramach tego badania zbierane i analizowane są poniższe dane:
- Kraj urodzenia
- Narodowość
- Płeć
- Wiek
- Języki ojczyste

Dane są przetwarzane przez:
- Głównego badacza (Magdalenę Borysiak) oraz jej promotorkę (Anne Abeillé)

Czy dane są przesyłane poza Unię Europejską: NIE

Poufność i bezpieczeństwo przechowywania danych osobowych są zapewnione w następujący sposób:
- Żadne dane osobowe zebrane przez zespół badawczy nie mogą zostać opublikowane, pozwalając w ten sposób na identyfikację uczestników badania;
- Dane osobowe uczestników zostaną zanonimizowane w ciągu 7 dni po ukończeniu zbierania danych.
- Dostęp do danych osobowych będzie:
    - Ograniczony – tylko zespół badawczy będzie miał dostęp do danych.
    - Chroniony identyfikatorem i hasłem.

Tylko dane które nie pozwalają na identyfikację uczestników zostaną opublikowane w formie pracy magisterskiej i/lub artykułu w czasopiśmie naukowym w celu poszerzania wiedzy naukowej. Wyniki badania mogą też zostać zaprezentowane na konferencji naukowej.

Ten dokument należy do Pana/i i może Pan/i podzielić się nim z rodziną i/lub przyjaciółmi w celu zasięgnięcia porady.

Dziękujemy za udział.

<u>Kontakt:</u>

Badaczka : Magdalena Borysiak
Adres mailowy: magdalena.borysiak@etu.u-paris.fr

Promotorka: Anne Abeillé
Adres mailowy: anne.abeille@u-paris.fr

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Figure 8: Information letter in English, the same for both experiments.

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INFORMATION LETTER REGARDING PARTICIPATION IN A STUDY

« Hybrid gender agreement pattern in Polish – experimental study »

Université Paris Cité, Paris Graduate School of Linguistics, Laboratoire de Linguistique Formelle

Dear Sir or Madam,

You are invited to participate in a study conducted by Magdalena Borysiak as part of their Master's thesis at the **Paris Graduate School of Linguistics / Laboratoire de Linguistique Formelle at Université Paris Cité**.

This document describes the study in which you will participate. It also answers any questions you may have based on the information currently available.

Before you decide whether to take part in the study or not, it is important that you understand the purpose of this study and what it entails.

I. Information about the study

1) Why are you being asked to participate in this study?

This study is offered to you because you fit the following criteria:

- You are at least 18 years old
- Polish is your first language
- You have a Polish nationality
- You currently live in Poland

2) What are the objectives of the study?

The objective of this study is to learn which kinds of sentences are acceptable to speakers of Polish.

3) What are the benefits of the study? Are there any special risks or constraints associated with your participation in the study?

The study will contribute to our understanding of how language works and how people use it. There are no risks associated with participating in this study beyond what you would encounter in your daily life. We will make sure that all participants are treated with respect and consideration.

4) What will you have to do during the study?

During the study you will see different sentences and you will have to judge how naturally they sound. If you are not sure what this means, try to imagine whether hearing those sentences in a conversation would be confusing for you. You will make your judgement using a 5-point scale that goes from "completely unacceptable" to "completely acceptable". At first you will have 3 practice questions so that you can become familiar with the task, after that you will proceed to the experiment. There will be 60 questions and you cannot go back to previous questions. There is no time constraint on any of the questions, but we encourage you to not to think too long about your answers and answer intuitively. The whole experiment should take you around 20 minutes.

5) What are your rights as a research participant?

You are free to accept or refuse to participate in this study without having to justify your decision, and without any consequences for you.
You can take the time you need to make your decision.
If you accept, you may interrupt your participation at any time, without giving a justification, by sending an email to: **magdalena.borysiak@etu.u-paris.fr.**

II. Information about the processing of your personal data

6) Will any personal data be processed in this study?

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Université
Paris Cité

Your participation implies the processing of personal data from the information about you that will be collected and produced in the context of this study.
The purpose of this processing is only scientific research. The legal basis is the performance of a task carried out in the public interest (art. 6.1.e of the General Data Protection Regulation).
The data controller is Université Paris Cité.

7) What are your rights with respect to your personal data?

You can access your data or request their deletion. You also have the right to object, the right to rectify and the right to limit the processing of your data. However, these rights may not be exercised after the full anonymization of your data.
To exercise these rights or ask questions about this research, you can contact the principal investigator of the study at magdalena.borysiak@etu.u-paris.fr**.** You will receive a reply as soon as possible and at the latest one month after your request was received.
In case of difficulty(s), you can also contact the Université Paris Cité Data Protection Officer at dpo@u-paris.fr.
If you believe, after having contacted the aforementioned people, that your « Informatique et Libertés » rights are not respected, you can send a complaint to the CNIL (www.cnil.fr).

8) How are your personal data managed?

As part of this study, the following information is collected and analysed:

- *Birth country*
- *Nationality*
- *Gender*
- *Age*
- *Native languages*

These data are processed by:

- The principal investigator (Magdalena Borysiak) and their research supervisor (Anne Abeillé)

Transfer of data outside of the European Union: **NO**

To ensure the security and confidentiality of your data, the following technical and organizational measures are in place:

- No personal data collected by the research team can be published or made public, which would allow the identification of the participants;
- The personal data of the participants will be anonymized maximum 7 days after data collection is finished.
- Access to personal data will be:
  - Limited – only the research team will have access to the data.
  - Protected by a personal identifier and a password

Only non-identifying data will be published in the form of a Master's thesis and/or an article in a scientific journal, in order to improve collective knowledge. The research results could also be presented at professional and scientific conferences.

This information letter is yours and you can share it with family members and/or friends for advice.

Thank you for your participation.

Contacts :

Researcher : Magdalena Borysiak
Email address : magdalena.borysiak@etu.u-paris.fr

Supervisor : Anne Abeillé
Email address : anne.abeille@u-paris.fr

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C Experimental, filler and control items from the experiments

Experimental items from Experiment 1:

1. **context** Grupa przestępcza składała się wyłącznie z kobiet.
   pre Najkrótszy wyrok otrzymał/otrzymała szpieg.
   post Szpieg otrzymał/otrzymała najkrótszy wyrok.
   pron Szpieg, który/która pozostaje w areszcie, spodziewa się najkrótszego wyroku.
   Q: Czy w zdaniu powiedziano jaki dokładnie to był wyrok? A: Nie

2. Przy stajni kręci się kilka pracowniczek, jedna z nich na koniu.
   Spośród nich najlepiej lekcje jazdy konnej prowadziła jeździec.
   Q: Czy w zdaniu była mowa o lekcjach jazdy konnej? A: Tak

3. Lekarki nie są zgodne co do diagnozy.
   Na wpływ leków uspokajających na przebieg choroby uwagę zwróciła psychiatra.
   Q: Czy w zdaniu była mowa o lekach uspokajających? A: Tak

4. Lekarki otworzyły salę i zaprowadziły pacjenta do stołu operacyjnego.
   Ponieważ wcześniej było to niemożliwe, wenflon pacjentowi założyła chirurg.
   Q: Czy w zdaniu był wspomniany wenflon? A: Tak

5. Na wydarzeniu pojawiły się również reprezentantki gminy Sterdyń.
   Najlepsze wrażenie na zgromadzonych zrobiła wójt.
   Q: Czy zdanie opisało na czym polegało wydarzenie? A: Nie

6. Islandia jest jednym z niewielu państw na świecie, gdzie kobiety pełnią funkcje prezydenta i szefa rządu.
   W 2024 roku swoje stanowisko zajęła premier.
   Q: Czy premier jest na stanowisku od 2009 roku? A: Nie

7. Na coroczne Dni Ekologii zaprosiłyśmy pracowniczki Instytutu Biologii Ssaków PAN.
   Wiekszość wykładów była prowadzona przez nasze koło naukowe, ale wykład o żubrach i turystyce poprowadziła naukowiec.
   Q: Czy w zdaniu był wspomniany wykład na temat dzików? A: Nie

8. Ostatnio kilka kobiet w jednostce wojskowej awansowało na wyższe stanowiska.
   Obowiązki głównego księgowego objęła porucznik.
   Q: Czy w zdaniu był wspomniany awans? A: Tak

9. Tegoroczne praktykantki w zakładach chemicznych, z których wszystkie poza jedną mają tytuł doktora, powoli zaczynają swoje praktyki.
   W zeszłym tygodniu praktyki zaczęła magister.
   Q: Czy w zdaniu były wspomniane praktyki? A: Tak

10. Wszystkie siostry przyjechały razem na miejsce samochodem.
    Ogromny kosz piknikowy z bagażnika wyciągnęła kierowca.
    Q: Czy w zdaniu był wspomniany kosz piknikowy? A: Tak

11. Ceremonię otworzyły najbardziej szanowane pracowniczki wydziału.
    Przemową skierowaną do pierwszorocznych studentów zaczęła dziekan.
    Q: Czy przemowa była skierowana do studentów pierwszego roku? A: Tak

12. Ósmego marca wszystkie pracowniczki gminy otrzymały po bukiecie kwiatów.
    Największy bukiet otrzymała burmistrz.
    Q: Czy prezentem wspomnianym w zdaniu były pralinki? A: Nie

13. Grupa policjantek weszła do knajpy.
    Spośród nich najwięcej jedzenia zamówiła sierżant.
    Q: Czy w zdaniu wspomniano jakie jedzenie zostało zamówione? A: Nie

14. Kobiety wśród władz uczelni bardzo poważnie potraktowały skargę.
    Postępowanie dyscyplinarne wszczęła rektor.
    Q: Czy w zdaniu mowa o postępowaniu dyscyplinarnym? A: Tak

15. W jednostce wojskowej, w której znajdują się same kobiety, odbył się ostatnio apel z okazji Święta Wojska Polskiego.
    Z tej okazji uroczystą zbiórkę poprowadziła pułkownik.
    Q: Czy w zdaniu wspomniania była uroczysta zbiórka wojskowa? A: Tak

16. Wszystkie najważniejsze kobiety z administracji miasta pojawiły się na otwarciu nowej biblioteki.
    O znaczeniu tego miejsca dla zapobieganiu samotności osób starszych mówiła prezydent.
    Q: Czy w zdaniu były wspomniane osoby starsze? A: Tak

17. W tym tygodniu odwiedzają nas mamy uczniów z naszej klasy, żeby opowiedzieć o swojej pracy.
    We wtorek przyszła architekt.
    Q: Czy dzień tygodnia, który był wspomniany w zdaniu, to czwartek? A: Nie

18. Delegacja z Ministerstwa Kultury i Dziedzictwa Narodowego, która odwiedziła Panteon w Paryżu, składała się wyłącznie z kobiet.
    Na grobie Marii Skłodowskiej-Curie kwiaty złożyła minister.
    Q: Czy w zdaniu mowa o zapaleniu znicza na grobie? A: Nie

19. Nad projektem miasta pracował zespół składający się wyłącznie z kobiet.
    W biurze przy Rynku Głównym pracowała inżynier.
    Q: Czy w zdaniu był wspomniany Rynek Główny? A: Tak

20. Dopiero lekarki ze Szpitala Bielańskiego potraktowały sprawę poważnie.
    Skierowanie na laparoskopię wystawiła ginekolog.
    Q: Czy w zdaniu była wspomniana gastroskopia? A: Nie

21. Przy przystanku pod teatrem stała grupa kobiet, spośród których jedna była wcześniej na widowni.
    Kiedy tylko autobus przyjechał, do środka wsiadła widz.
    Q: Czy w zdaniu wspomniany był tramwaj? A: Nie

22. Grupa studentek przyszła z prośbą o przesunięcie terminu kolokwium.
    O odwołanych ćwiczeniach i konieczności odrobienia ich przed kolokwium przypomniała starosta.
    Q: Czy studentki w zdaniu chcą odwołać ćwiczenia? A: Nie

23. Na spotkanie przyszła grupa starszych kobiet, z których jedna brała udział w Powstaniu Warszawskim, żeby opowiedzieć o życiu w Polsce w czasie II wojny światowej.
    Na samym początku głos zabrała powstaniec.
    Q: Czy w zdaniu była mowa o osobie która brała udział w Powstaniu Warszawskim? A: Tak

24. Wczoraj po południu odbył się panel, w którym kobiety opowiadały o swoich doświadczeniach w zawodach zdominowanych przez mężczyzn.
    O przemocy ze względu na płeć opowiadała polityk.
    Q: Czy w zdaniu wspomniano o przemocy na tle rasowym? A: Nie

Experimental items from Experiment 2:

1. **pre** Na rowerach przyjechały/przyjechali chłopaki.
   post Chłopaki przyjechały/przyjechali na rowerach.
   pron Chłopaki, które/którzy jadą na rowerach, mają szansę przyjechać na czas.
   Q: Czy w zdaniu były wspomniane rowery? A: Tak

2. Wtedy gołębie i króliki chodowały górniki.
   Q: Czy w zdaniu była mowa o papugach? A: Nie

3. W tym hotelu mieszkały generały.
   Q: Czy w zdaniu była mowa o hotelu? A: Tak

4. Zakupu tych opon mi odradziły mechaniki.
   Q: Czy w zdaniu była mowa o oponach? A: Tak

5. Podczas przerw między rundami graczom źle doradzały trenery.
   Q: Czy w zdaniu była mowa o przeprowadzce? A: Nie

6. Na forum ekonomiczne przyjechały prezydenty.
   Q: Czy w zdaniu wspomniano forum ekonomiczne? A: Tak

7. Z naszej parafii na misje pojechały księdze.
   Q: Czy w zdaniu była mowa o chórze? A: Nie

8. Największy stół położony przy oknach zajęły proboszcze.
   Q: Czy stół znajdował się przy drzwiach? A: Nie

9. Z nożycami do rozcinania auta przyjechały strażaki.
   Q: Czy w zdaniu była mowa o rozcinaniu samochodu? A: Tak

10. Na salę operacyjną weszły chirurgi.
    Q: Czy w zdaniu wspomniano o pielęgniarkach? A: Nie

11. Na samym końcu do samolotu weszły piloty.
    Q: Czy w zdaniu była mowa o stewardessach? A: Nie

12. Zazwyczaj z potłuczonymi nosami do nas przychodziły boksery.
    Q: Czy w zdaniu wspomniano o urazie kręgosłupa? A: Nie

13. Większe dotacje na organizację festiwalu wywalczyły dyrektory.
    Q: Czy w zdaniu była mowa o organizacji festiwalu? A: Tak

14. W trójkąt stojący sto metrów za naszym samochodem wjechały policjanty.
    Q: Czy w zdaniu była mowa o straży pożarnej? A: Nie

15. Przez hałas na ulicy z zakładu wyszły rzeźniki.
    Q: Czy w zdaniu wspomniano o rzeźnikach? A: Tak

16. Przed wejściem do zbiornika wodnego nas zatrzymały strażniki.
    Q: Czy w zdaniu wspomniano o zbiorniku wodnym? A: Tak

17. Proszku na szczury w piwnicy podsypały dozorce.
    Q: Czy w zdaniu wspomniano o parterze budynku? A: Nie

18. Klimatyzowanymi limuzynami jeździły prezesy.
    Q: Czy w zdaniu była mowa o limuzynach? A: Tak

19. Na nagrania do Ameryki Południowej poleciały reżysery.
    Q: Czy w zdaniu wspomniano o Azji? A: Nie

20. Ze sprzedaży swoich produktów na miejskich targowiskach żyły rolniki.
    Q: Czy w zdaniu była mowa o targowiskach miejskich? A: Tak

21. Blokujący wodę wał postawiły inżyniery.
    Q: Czy w zdaniu wspomniano o moście? A: Nie

22. W tym budynku pracowały szewce.
    Q: Czy w zdaniu była mowa o szewcach? A: Tak

23. Największy problem ze znalezieniem mojego domu miały kuriery.
    Q: Czy zdanie opisywało zwrot nadanej przesyłki? A: Nie

24. Dużo kłopotu z naszymi bagażami miały celniki.
    Q: Czy w zdaniu była mowa o bagażach? A: Tak

Control items:

1. **context** Nie wszystkie dzieci chciałyby mieć wieczne wakacje.
   correct Syn sąsiadki jest szczęśliwy że idzie do szkoły.
   incorrect Syn sąsiadki jest szczęśliwa że idzie do szkoły.
   Q: Czy syn sąsiadki chodzi do szkoły? A: Tak

2. Moja córka wychowuje dzieci sama.
   Mąż mojej córki jest szczęśliwy w Hollywood.
   Q: Czy w zdaniu wspomniano Hollywood? A: Tak

3. Kiedy wróciłem z pracy, na podjeździe stało jego auto.
   Kochanek mojej żony został przyłapany.
   Q: Czy według zdania żona miała kochanka? A: Tak

4. Na opiekunkę możemy wziąć kogoś z rodziny.
   Kuzynka mojego ojca jest miła dla dzieci.
   Q: Czy według zdania ojciec ma kuzynkę? A: Tak

5. Widzę coraz więcej starszych mężczyzn spotykających się z dwudziestolatkami.
   Partnerka mojego wujka jest młoda.
   Q: Czy wspomniana w zdaniu kobieta jest młoda? A: Tak

6. Po drodze do Hiszpanii możemy się zatrzymywać u znajomych.
   Córka mojego przyjaciela wyjechała do Niemiec.
   Q: Czy w zdaniu wspomniano o Niemczech? A: Tak

7. Radio w samochodzie mi nie działa.
   Czubek anteny zahaczył o sufit w tunelu.
   Q: Czy według zdania samochód jechał przez most? A: Nie

8. Jeszcze nie skończyłem sprzątać.
   Fragment posadzki jest poplamiony.
   Q: Czy posadzka jest czysta? A: Nie

9. Źle mi się spało dzisiaj w nocy.
   Blask latarni mnie budził.
   Q: Czy powodem budzenia był hałas u sąsiadów? A: Nie

10. Usłyszałam hałas w kuchni.
    Pokrywka garnka spadła na ziemię.
    Q: Czy słoik upadł na ziemię? A: Nie

11. Spodobała mi się taka jedna piosenka, ale nie mogę sobie przypomnieć tytułu ani nazwy zespołu.
    Okładka albumu miała zdjęcie kwiatów.
    Q: Czy według zdania na okładce była twarz? A: Nie

12. To miejsce nie zachęca klientów do wejścia do środka.
    Witryna sklepu wygląda na starą.
    Q: Czy zdanie wspomina o szyldzie sklepu? A: Nie

Filler items:

1. **context** Bardzo lubię zajmować się wystrojem wnętrz.
   item Mój brat kupił mieszkanie, które kazałem mu wyremontować.
   Q: Czy brat kupił dom? A: Nie

2. Pracownicy sklepu niezbyt mi pomogli z usterką.
   Otrzymałem instrukcję obsługi, którą powiedziano mi żebym przeczytał.
   Q: Czy w zdaniu mowa o wierszu? A: Nie

3. Jutro przyjeżdża do mnie rodzina i chcę ich dobrze ugościć.
   Na obiad zaplanowałem danie rybne, które potrafię przygotować.
   Q: Czy danie rybne ma być na kolację? A: Nie

4. Po tym kursie bardzo dobrze znam język grecki.
   Przeczytałem grecką powieść i zastanawiam się, jak ją przetłumaczyć.
   Q: Czy w zdaniu wspomniano o niemieckiej powieści? A: Nie

5. Szukam agencji opiekunek do dzieci.
   Mam córkę i zastanawiam się, gdzie znaleźć dla niej opiekę w środy.
   Q: Czy córka potrzebuje opieki w piątki? A: Nie

6. Muszę kupić otwieracz do wina.
   Kupiłem butelkę i nie wiem, jak ją otworzyć.
   Q: Czy w zdaniu mowa o otwieraniu drzwi? A: Nie

7. Jutro chcę zadzwonić do brata.
   Mam problem, którego nie potrafię rozwiązać samodzielnie.
   Q: Czy osoba w zdaniu może rozwiązać problem samodzielnie? A: Nie

8. Strona internetowa sklepu jest trudna do nawigowania.
   Potrzebuję odpowiedzi i chciałbym ją znaleźć na tej stronie.
   Q: Czy osoba w zdaniu zna odpowiedź? A: Nie

9. Szukam zeszytu w którym babcia zapisywała hasła.
   Znalazłem na strychu sejf i nie wiem jak go otworzyć.
   Q: Czy według zdania na strychu była skrzynia? A: Nie

10. Potrzebuję jakiejś okazji do świętowania.
    Kupiłem drogie wino, którego nie wiem kiedy spróbować.
    Q: Czy wino było tanie? A: Nie

11. Opieka nad psem nie byłaby zbyt dużą odpowiedzialnością dla mnie.
    Mam w domu dużo roślin, które umiem pielęgnować.
    Q: Czy w domu jest mało roślin? A: Nie

12. Jak będę w centrum miasta to wejdę do apteki.
    Przepisano mi nowe tabletki, które nie wiem gdzie kupić.
    Q: Czy przepisano nową maść? A: Nie

13. W weekend planuję zrobić przemeblowanie.
    Mam nowe biurko i szukam miejsca, gdzie je postawię w swoim domu.
    Q: Czy w zdaniu wspomniano o biurku? A: Tak

14. Muszę zadzwonić do sklepu.
    Dostałem regał na książki, ale nie wiem jak go złożyć w domu.
    Q: Czy w zdaniu wspomniano o regale na książki? A: Tak

15. Ta strona jest zbyt skomplikowana jak dla mnie.
    Mam pytanie i nie wiem, gdzie je umieścić na tym forum.
    Q: Czy osoba w zdaniu ma pytanie? A: Tak

16. Bardzo mi się podobało na zajęciach z ogrodnictwa.
    Wymieniłem się nasionami i planuję je zasadzić w ogrodzie.
    Q: Czy osoba w zdaniu chce zasadzić nasiona w ogrodzie? A: Tak

17. Myślę, że możemy zakończyć na dziś.
    Pozostała jedna kwestia i planuję poruszyć ją następnym razem.
    Q: Czy pozostała kwestia będzie poruszona następnym razem? A: Tak

18. Zawsze wolałem psy, ale ostatnio się przekonuję do kotów.
    Mój sąsiad powierzył mi kotka i powiedział, jak się nim opiekować.
    Q: Czy kot był powierzony przez sąsiada? A: Tak

19. Rodzice pozwolili mi wracać samemu do domu ze szkoły w pierwszej klasie.
    Powierzono mi klucze i pamiętam, że odłożyłem je w bezpieczne miejsce.
    Q: Czy klucze są w bezpiecznym miejscu? A: Tak

20. Chciałbym mieć trochę spokoju w domu w sierpniu.
    Mam syna i rozważam wysłanie go na kolonie.
    Q: Czy w zdaniu wspomniany jest syn? A: Tak

21. Ostatnio mam więcej czasu na moje hobby.
    Skończyłem kolejną rzeźbę i zastanawiam się gdzie ją postawić.
    Q: Czy w zdaniu wspomniano o rzeźbie? A: Tak

22. Zanim zaczniemy gotować, pojadę na szybkie zakupy.
    Brakuje mi jednego składnika i wiem gdzie go kupić.
    Q: Czy osoba w zdaniu wie gdzie kupić brakujący składnik? A: Tak

23. Chcę spać latem na działce, ale jest pełno komarów.
    Kupiłem siatkę na komary, którą wiem jak zamontować.
    Q: Czy w zdaniu mowa o siatce na komary? A: Tak

24. Czekam na jakieś zniżki na loty do Włoch.
    Mam kilka dni urlopu i zastanawiam się kiedy je wykorzystać.
    Q: Czy w zdaniu wspomniano o urlopie? A: Tak

D Coefficients from statistical modeling of the experimental data

	Estimate	Std. error	Z value	Pr(>|z|)
Profession noun	3.7672	0.4546	8.287	<2e-16 ***
Semantic	0.5275	0.3156	1.671	0.0946 .
Controller first	-0.1432	0.3301	-0.434	0.6644
Relative pronoun	0.5353	0.3190	1.678	0.0933 .
Profession noun:Semantic	-0.6341	0.4105	-1.545	0.1224
Profession noun:Controller first	0.1067	0.4231	0.252	0.8009
Profession noun:Relative pronoun	0.1422	0.4161	0.342	0.7326
Semantic:Controller first	0.3750	0.4478	0.837	0.4023
Semantic:Relative pronoun	-0.7160	0.4327	-1.655	0.0980 .
Profession noun:Semantic:Controller first	0.2532	0.5841	0.433	0.6647
Profession noun:Semantic:Relative pronoun	0.4616	0.5768	0.800	0.4235

Table 8: Coefficients of the mixed effects ordinal regression model using data from both experiments.
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Semantic	-0.09523	0.25584	-0.372	0.7097
Controller first	-0.03820	0.25593	-0.149	0.8814
Relative pronoun	0.63759	0.25970	2.455	0.0141 *
Semantic:Controller first	0.60186	0.36529	1.648	0.0994 .
Semantic:Relative pronoun	-0.25598	0.37176	-0.689	0.4911

Table 9: Coefficients of the mixed effects ordinal regression model using data from Experiment 1.
Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1.

	Estimate	Std. error	Z value	Pr(>|z|)
Semantic	0.5545	0.3259	1.701	0.0889 .
Controller first	-0.1881	0.3399	-0.554	0.5799
Relative pronoun	0.6164	0.3289	1.874	0.0610 .
Semantic:Controller first	0.4655	0.4652	1.001	0.3170
Semantic:Relative pronoun	-0.8465	0.4512	-1.876	0.0606 .

Table 10: Coefficients of the mixed effects ordinal regression model using data from Experiment 2.
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.